

The Baker–Campbell–Hausdorff Formula

Complete Expansions, Proofs, and Analytic Qualifications

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Abstract

We give a self-contained derivation of the mathematical formulas and explicitly stated results in the Wikipedia article on the Baker–Campbell–Hausdorff formula, with the hypotheses needed to make each assertion correct. The main algebraic proof establishes the primitive-element theorem and the Dynkin–Specht–Wever projection, and then derives a finite coefficient formula for every associative word and Dynkin’s complete commutator expansion. A second, differential route gives the Poincaré integral, Bernoulli recurrences, a complete expansion in the number of occurrences of one variable, and an explicit bivariate generating function for its quadratic part. We prove local convergence with quantitative remainder bounds, classify the logarithms in the article’s divergence example, derive global identities for central and eigenvector commutators, and construct every Zassenhaus exponent with a convergent-product majorant. Trotter, symmetric, and recursively higher-order product formulas follow. The infinitesimal formulas are developed into all-order expressions for coframes, pullback metrics, and volume densities. Finally, the Weyl and displacement identities are proved in concrete Hilbert-space representations, rather than inferred from an unjustified manipulation of unbounded operators. Exact rational code and coefficient tables through degree twelve accompany the paper.

Keywords. Free Lie algebra; noncommutative power series; BCH; Dynkin projection; Bernoulli numbers; Zassenhaus factorization; exponential coordinates; Weyl relations.

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1 Scope, conventions, and the three meanings of BCH

The equation

$$e^X e^Y = e^Z \quad (1.1)$$

is not, by itself, a definition of a unique logarithm. The exponential can fail to be injective, and a product of exponentials need not be an exponential *in the specified Lie algebra*. What is universal is the formal series

$$Z(X, Y) = \text{BCH}(X, Y) = \log(e^X e^Y) = \sum_{n \geq 1} Z_n(X, Y), \quad (1.2)$$

where Z_n has total degree n . A principal matrix logarithm, a continued branch of a logarithm, and a convergent BCH series are related but distinct objects.

The coverage target is the version of the source article identified by revision number 1368909113, consulted on 16 September 2026 [1]. Every displayed mathematical identity and every mathematical assertion stated in its main text or mathematical notes is addressed below. Historical priority statements, application surveys, and theorems merely named in links are not additional proof obligations; in particular, the article does not state the Stone–von Neumann or Golden–Thompson theorems. [Section C](#) gives a claim-by-claim concordance. Several assertions about geometry require correction, not proof as written: ad_X is generally singular and is not the coframe matrix; a Killing form need not be a metric; and an arbitrary pullback can be degenerate. We give the correct statements and explicit counterexamples.

1.1 Algebraic and analytic settings

Throughout the formal arguments, $\mathbb{K}$ is a field of characteristic zero. Let

$$\mathcal{T} = \mathbb{K}\langle X, Y \rangle = \bigoplus_{n \geq 0} \mathcal{T}_n, \quad \widehat{\mathcal{T}} = \prod_{n \geq 0} \mathcal{T}_n,$$

where $\mathcal{T}_n$ has as a basis the words of length n in X, Y . The empty word is the unit 1. Multiplication is concatenation, extended degree by degree. Thus the coefficient of a word in a product is always a finite sum. Write

$$F^r \widehat{\mathcal{T}} = \prod_{n \geq r} \mathcal{T}_n.$$

A sum converges *formally* when every fixed degree eventually stabilizes. No norm or numerical limiting process is involved. The bracket is

$$[A, B] = AB - BA, \quad \text{ad}_A B = [A, B]. \quad (1.3)$$

Associativity gives antisymmetry and the Jacobi identity by expansion of the twelve resulting products. In particular,

$$[A, [B, C]] = [[A, B], C] + [B, [A, C]], \quad [\text{ad}_A, \text{ad}_B] = \text{ad}_{[A, B]}. \quad (1.4)$$

The Lie subalgebra of $\mathcal{T}$ generated by X, Y is denoted $\mathcal{L}(X, Y)$; its degree completion is $\widehat{\mathcal{L}}(X, Y)$.

For analytic statements, $\mathcal{A}$ is a real or complex unital Banach algebra with a submultiplicative norm. Finite matrices are the main example. When discussing arbitrary finite-dimensional Lie groups we use their ordinary smooth exponential map and derive the same formulas intrinsically. Unbounded Hilbert-space operators are handled separately in [section 11](#).

Lemma 1.1 (Formal exponential and logarithm). *The series*

$$\exp A = \sum_{n \geq 0} \frac{A^n}{n!}, \quad \log(1 + U) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} U^n \quad (1.5)$$

are mutually inverse bijections between $F^1\widehat{\mathcal{T}}$ and $1 + F^1\widehat{\mathcal{T}}$. If A, B commute, then $e^{A+B} = e^A e^B$; if $g, h \in 1 + F^1\widehat{\mathcal{T}}$ commute, then $\log(gh) = \log g + \log h$.

Proof. Positive order makes every coefficient in each substitution a finite sum. The scalar identities $\log(\exp t) = t$ and $\exp(\log(1 + t)) = 1 + t$ follow, for example, by differentiating the corresponding formal scalar series and checking the constant terms. Substitution of the single element A or U is legitimate: its powers commute with each other even though the ambient algebra need not be commutative. For commuting A, B , the binomial theorem gives

$$\sum_{n \geq 0} \frac{(A + B)^n}{n!} = \sum_{r, s \geq 0} \frac{A^r B^s}{r! s!} = e^A e^B.$$

When g, h commute, their logarithms commute as well, so the preceding identity and the inverse property give the last assertion. $\square$

In a Banach algebra the exponential series converges for every A , since $\|A^n/n!\| \leq \|A\|^n/n!$. The logarithm series converges whenever $\|U\| < 1$. Formal identities involving absolutely convergent substitutions remain valid analytically.

1.2 How a universal series is evaluated

A homogeneous Lie polynomial can be evaluated at any pair of elements of any characteristic-zero Lie algebra. The identification of $\mathcal{L}(X, Y)$ with the abstract free Lie algebra, and the fact that the free associative algebra is its universal enveloping algebra, are proved in [section A](#). For a general Lie algebra without a topology, the unambiguous statement is

$$\text{BCH}(tX, tY) = \sum_{n \geq 1} t^n Z_n(X, Y) \in t\mathfrak{g}[[t]],$$

not that the sum at $t = 1$ is an element of $\mathfrak{g}$. In a complete positively filtered Lie algebra it can instead be evaluated by filtration convergence. In a nilpotent Lie algebra it is a finite polynomial.

Our convention for an iterated commutator is *right nesting*:

$$R(a_1 \cdots a_n) = [a_1, [a_2, \dots, [a_{n-1}, a_n] \dots]], \quad R(a_1) = a_1. \quad (1.6)$$

The letters may be X or Y . Thus $R(XXY) = \text{ad}_X^2 Y$ and $R(YYX) = \text{ad}_Y^2 X$. The symbols $X^r Y^s$ inside R describe a word, not a commutator of associative powers.

2 Why the logarithm is a Lie series

We prove the primitive-element statement used in the source article rather than treating it as a black box. The proof simultaneously constructs the projection that produces Dynkin's formula.

2.1 The coproduct and group-like elements

Define the continuous algebra homomorphism

$$\Delta : \widehat{\mathcal{T}} \longrightarrow \widehat{\mathcal{T}} \widehat{\otimes} \widehat{\mathcal{T}}, \quad \Delta X = X \otimes 1 + 1 \otimes X, \quad \Delta Y = Y \otimes 1 + 1 \otimes Y. \quad (2.1)$$

Here the target is the *completed* tensor product $\prod_{p,q \geq 0} \mathcal{T}_p \otimes \mathcal{T}_q$, with the total-degree interpretation of multiplication and infinite sums. An ordinary algebraic tensor product is not sufficient for arbitrary formal series. Explicitly, if $w = a_1 \cdots a_n$, then

$$\Delta w = \sum_{I \subseteq \{1, \dots, n\}} a_I \otimes a_{I^c}, \quad (2.2)$$

where a_I lists the selected letters in their original order. The counit ε selects the constant coefficient. Coassociativity and the counit equations can be checked on X, Y and then extended by multiplicativity and continuity.

An element P is *primitive* when $\Delta P = P \otimes 1 + 1 \otimes P$. An element $g \in 1 + F^1 \widehat{\mathcal{T}}$ is *group-like* when $\Delta g = g \otimes g$. The normalization of the constant term is part of this definition.

Proposition 2.1. *For $P \in F^1 \widehat{\mathcal{T}}$, P is primitive if and only if e^P is group-like. Group-like elements form a group under multiplication.*

Proof. Continuity and multiplicativity give $\Delta(e^P) = e^{\Delta P}$. If P is primitive, the two tensor factors commute, so

$$e^{\Delta P} = e^{P \otimes 1} e^{1 \otimes P} = e^P \otimes e^P.$$

Conversely, if $g = e^P$ is group-like, then

$$\begin{aligned} \Delta P &= \log(\Delta g) = \log((g \otimes 1)(1 \otimes g)) \\ &= \log(g \otimes 1) + \log(1 \otimes g) = P \otimes 1 + 1 \otimes P. \end{aligned}$$

The commuting-logarithm rule applies because both factors have constant term one. Finally, $\Delta(gh) = (g \otimes g)(h \otimes h) = gh \otimes gh$, and $\Delta(g^{-1}) = (\Delta g)^{-1} = g^{-1} \otimes g^{-1}$. $\square$

2.2 An explicit primitive-element projection

For a word w of length n , define

$$S(w) = (-1)^n \text{rev}(w), \quad N(w) = nw, \quad S(1) = 1, \quad N(1) = 0.$$

The map S is an antihomomorphism. For endomorphisms f, g of $\mathcal{T}$, their convolution is $f * g = m(f \otimes g)\Delta$, where m is multiplication.

Lemma 2.2 (Antipode and right Dynkin operator). *The map S satisfies $\text{Id} * S = S * \text{Id} = \eta \varepsilon$, where η inserts scalars. The convolution*

$$D = N * S \quad (2.3)$$

satisfies $D(w) = R(w)$ for every nonempty word.

Proof. Use the finite notation $\Delta w = \sum u \otimes v$ and put $K(w) = \sum uS(v)$. If a is a letter, then $\Delta(aw) = \sum au \otimes v + u \otimes av$ and $S(av) = -S(v)a$. Consequently

$$K(aw) = aK(w) - K(w)a = 0.$$

Since $K(1) = 1$, this proves $\text{Id} * S = \eta\varepsilon$. The corresponding calculation using a final letter proves the other antipode identity (or one can use uniqueness of the convolution inverse in the connected graded algebra).

Next, $N(au) = aN(u) + au$. Therefore

$$\begin{aligned} D(aw) &= \sum N(au)S(v) + N(u)S(av) \\ &= aD(w) - D(w)a + aK(w). \end{aligned}$$

For $w \neq 1$, the last term vanishes. For $w = 1$, it equals a , so $D(a) = a$. Induction now gives $D(aw) = [a, D(w)]$, which is exactly right nesting. $\square$

Theorem 2.3 (Primitive elements and Dynkin–Specht–Wever). *A homogeneous $P \in \mathcal{T}_n$, $n > 0$, is primitive if and only if it belongs to $\mathcal{L}(X, Y)_n$. For such a P ,*

$$D(P) = nP, \quad P = \frac{1}{n} \sum_{|w|=n} [w]P R(w). \quad (2.4)$$

Thus D/n is an idempotent projection of $\mathcal{T}_n$ onto $\mathcal{L}(X, Y)_n$, and the primitive elements of $\widehat{\mathcal{T}}$ are exactly $\widehat{\mathcal{L}}(X, Y)$.

Proof. If P is homogeneous and primitive, then

$$D(P) = m(N \otimes S)(P \otimes 1 + 1 \otimes P) = nP.$$

By [theorem 2.2](#), $D(P)$ is a linear combination of nested commutators. Division by n is permitted in characteristic zero, proving $P \in \mathcal{L}$ and the displayed formula.

Conversely, X, Y are primitive, and the bracket of primitive elements is primitive: the mixed terms in $[\Delta U, \Delta V]$ cancel because the tensor factors commute. Hence every Lie polynomial is primitive. Since D always takes values in Lie polynomials, applying D once more gives $D^2 = nD$ on degree n . Finally, Δ preserves total degree, so a completed series is primitive exactly when each homogeneous component is primitive. Its degree-zero component must be zero, since a primitive scalar c would satisfy $c = 2c$. $\square$

Theorem 2.4 (Formal BCH theorem). *There is a unique $Z \in F^1\widehat{\mathcal{T}}$ satisfying $e^Z = e^Xe^Y$. It belongs to $\widehat{\mathcal{L}}(X, Y)$, has rational coefficients, and its degree-one term is $X + Y$.*

Proof. Uniqueness and existence in the completed associative algebra follow from [theorem 1.1](#). Since X, Y are primitive, their exponentials and the product e^Xe^Y are group-like. Its logarithm is primitive by [theorem 2.1](#), and hence a Lie series by [theorem 2.3](#). Rationality follows from the factorial and logarithmic coefficients. Taking degree one gives $Z_1 = X + Y$. $\square$

This proves all the structural assertions about primitive and group-like elements needed for BCH. The Hopf-algebra interpretation is compatible with universal enveloping algebras, as explained and proved in [section A](#).

3 Complete coefficient formulas and the first six degrees

The structural theorem becomes completely explicit by applying D to the ordinary logarithm. Dynkin's formula is historically attributed to [2]; modern computational treatments include [4]. The proof below derives its coefficients directly.

3.1 Associative blocks and a finite formula for every word

Set

$$U = e^X e^Y - 1 = \sum_{\substack{r,s \geq 0 \\ r+s > 0}} \frac{X^r Y^s}{r!s!}. \quad (3.1)$$

Substituting this into the logarithm gives the full associative expansion

$$Z = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{X^{r_1} Y^{s_1} \cdots X^{r_k} Y^{s_k}}{\prod_{i=1}^k r_i! s_i!}. \quad (3.2)$$

The inner sum is not a sum over independent commutators; at this stage it is simply noncommutative multiplication.

For a nonempty word v , define

$$a(v) = \begin{cases} 1/(r!s!), & v = X^r Y^s, \quad r + s > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (3.3)$$

A cut $w = w_1 \cdots w_k$ always means consecutive, nonempty subwords.

Theorem 3.1 (All associative coefficients). *For every word w of length $n > 0$,*

$$c(w) := [w]Z = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{w=w_1 \cdots w_k} \prod_{i=1}^k a(w_i). \quad (3.4)$$

This is a finite, nonrecursive formula for every coefficient at every order. Equivalently,

$$Z_n = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i + s_i > 0 \\ \sum_i (r_i + s_i) = n}} \frac{X^{r_1} Y^{s_1} \cdots X^{r_k} Y^{s_k}}{\prod_i r_i! s_i!}. \quad (3.5)$$

Proof. The coefficient of w in U^k is the sum over all its cuts into k nonempty pieces, with the product of their coefficients in U . Those coefficients are precisely $a(w_i)$. There are no contributions for $k > n$. Substitution in $\log(1 + U)$ proves both formulas. $\square$

For example, for $w = XXY$ the one-piece contribution is $1/2$, the two-piece contribution is $-(1 + 1/2)/2 = -3/4$, and the three-piece contribution is $1/3$. Their sum is $1/12$. For $w = YXX$, the one-piece contribution is zero, the two-piece contribution is $-1/2$, and the three-piece contribution is $1/3$, giving $-1/6$.

3.2 Dynkin’s complete commutator formula

Theorem 3.2 (Dynkin formula). *With $N = \sum_{i=1}^k (r_i + s_i)$,*

$$Z = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0}} \frac{R(X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k})}{N \prod_i r_i! s_i!}. \quad (3.6)$$

Equivalently,

$$Z_n = \frac{1}{n} \sum_{|w|=n} c(w) R(w). \quad (3.7)$$

A summand in [equation \(3.6\)](#) vanishes when $s_k > 1$, or when $s_k = 0$ and $r_k > 1$.

Proof. Each Z_n is a homogeneous Lie polynomial. Apply [equation \(2.4\)](#) to [equation \(3.5\)](#); the operator replaces every word by its right-nested bracket and division by its length gives the factor $1/N$. Summing over degrees gives the formula. Under either stated condition the two innermost letters coincide, so the innermost bracket is zero. $\square$

At every fixed degree the sums are finite. Analytic regrouping of all degrees requires a convergence argument; explicit sufficient conditions are proved in [section 6](#).

Corollary 3.3 (Every nonlinear term contains the basic commutator). *For $n \geq 2$, Z_n belongs to the Lie ideal generated by $[X, Y]$. In particular it vanishes when $[X, Y] = 0$.*

Proof. In the quotient by that ideal, the two generators commute. Every bracket of length at least two in those generators is consequently zero, so the quotient Lie algebra is abelian. The image of Z is $X + Y$ by the commuting exponential identity. Homogeneous components of higher degree therefore lie in the ideal. $\square$

The phrase “contains $[X, Y]$ ” is an ideal-theoretic statement; it does not mean that an arbitrary chosen display must visibly have the same innermost bracket in every term.

3.3 Symmetries, including a useful coefficient check

Proposition 3.4. *The formal BCH series satisfies*

$$Z(Y, X) = -Z(-X, -Y), \quad (3.8)$$

$$Z_n(Y, X) = (-1)^{n+1} Z_n(X, Y), \quad (3.9)$$

$$\text{rev}(Z_n(X, Y)) = (-1)^{n+1} Z_n(X, Y). \quad (3.10)$$

In particular, coefficients transform by the same sign under either swapping X, Y or reversing every word.

Proof. Since $e^{-X}e^{-Y} = (e^Y e^X)^{-1}$, taking the formal logarithm gives [equation \(3.8\)](#). Taking homogeneous components gives [equation \(3.9\)](#). Word reversal is an antihomomorphism, so it takes $e^X e^Y$ to $e^Y e^X$ and commutes with the single-element logarithm series. This gives [equation \(3.10\)](#). $\square$

3.4 Every term through degree six

With the right-nesting convention [equation \(1.6\)](#), the full answer through degree six is

$$Z_1 = X + Y, \quad (3.11)$$

$$Z_2 = \frac{1}{2}R(XY), \quad (3.12)$$

$$Z_3 = \frac{1}{12}(R(XXY) + R(YYX)), \quad (3.13)$$

$$Z_4 = -\frac{1}{24}R(YXXY), \quad (3.14)$$

$$\begin{aligned} Z_5 = & -\frac{1}{720}(R(YYYYX) + R(XXXXY)) \\ & + \frac{1}{360}(R(XYYYX) + R(YXXXY)) \\ & + \frac{1}{120}(R(YXYXY) + R(XYXXY)), \end{aligned} \quad (3.15)$$

$$\begin{aligned} Z_6 = & \frac{1}{240}R(XYXYXY) + \frac{1}{720}(R(XYXXXXY) - R(XXYYXY)) \\ & + \frac{1}{1440}(R(XYYYXY) - R(XXYXXY)). \end{aligned} \quad (3.16)$$

There is no omitted term of any of these degrees. Equations [equations \(3.4\)](#) and [\(3.6\)](#) define every subsequent degree, rather than leaving the ellipsis unspecified.

For direct comparison with associative matrix multiplication,

$$Z_2 = \frac{1}{2}(XY - YX), \quad (3.17)$$

$$Z_3 = \frac{1}{12}(X^2Y + XY^2 - 2XYX + Y^2X + YX^2 - 2YXY), \quad (3.18)$$

$$Z_4 = \frac{1}{24}(X^2Y^2 - 2XYXY - Y^2X^2 + 2YXYX). \quad (3.19)$$

For instance,

$$[X, [X, Y]] = X^2Y - 2XYX + YX^2, \quad [Y, [Y, X]] = Y^2X - 2YXY + XY^2,$$

which proves the cubic conversion. Expanding $-[Y, [X, [X, Y]]]/24$ proves [equation \(3.19\)](#).

Here is a fully specified finite verification of the fifth- and sixth-degree displays, also usable for any larger display. For $w = a_1 \cdots a_n$,

$$R(w) = \sum_{S \subseteq \{1, \dots, n-1\}} (-1)^{|S|} a_{i_1} \cdots a_{i_{n-1-|S|}} a_n a_{j_{|S|}} \cdots a_{j_1}, \quad (3.20)$$

where $i_1 < \cdots < i_{n-1-|S|}$ enumerate S^c and $j_1 < \cdots < j_{|S|}$ enumerate S . At each bracket one either places the outer letter at the left or places it at the right with a minus sign, proving this identity by induction. Substitute [equation \(3.20\)](#) in [equations \(3.15\)](#) and [\(3.16\)](#) and compare every word coefficient with the finite cut formula [equation \(3.4\)](#). The complete nonzero coefficient certificates are printed in [section B](#); all remaining coefficients are zero in both calculations. This proves the displayed formulas by finite rational arithmetic, without choosing a commutator basis or appealing to a numerical approximation.

4 Conjugation, differentiation, and the integral formula

This section gives a second route to BCH. It also proves the Campbell identity and every infinitesimal identity used later. Parameter differentiation of exponentials is treated systematically in [\[7\]](#); all formulas required here are derived below.

4.1 The Campbell conjugation identity

Theorem 4.1 (Conjugation by an exponential). *In the formal setting, and for all X, Y in a Banach algebra,*

$$e^{sX} Y e^{-sX} = e^{s \operatorname{ad}_X} Y = \sum_{n \geq 0} \frac{s^n}{n!} \operatorname{ad}_X^n Y. \quad (4.1)$$

For matrices this says $\operatorname{Ad}_{e^X} = e^{\operatorname{ad}_X}$, where $\operatorname{Ad}_A Y = A Y A^{-1}$. Consequently,

$$e^X e^Y e^{-X} = \exp(e^{\operatorname{ad}_X} Y), \quad (4.2)$$

$$e^X e^Y = \exp\left(\sum_{n \geq 0} \frac{\operatorname{ad}_X^n Y}{n!}\right) e^X. \quad (4.3)$$

Proof. Let $F(s) = e^{sX} Y e^{-sX}$. Differentiation gives

$$F'(s) = X F(s) - F(s) X = \operatorname{ad}_X F(s), \quad F(0) = Y.$$

The unique formal solution, obtained by comparing powers of s , is the series in [equation \(4.1\)](#). In the Banach setting it is the unique solution of a bounded linear differential equation, and

$$\sum_{n \geq 0} \frac{|s|^n}{n!} \|\operatorname{ad}_X^n Y\| \leq e^{2|s| \|X\|} \|Y\|.$$

Thus it converges without a smallness restriction. Conjugation preserves every power of Y , so it preserves its exponential series. This proves [equation \(4.2\)](#); multiplying on the right by e^X proves [equation \(4.3\)](#). $\square$

A useful purely associative version is

$$\operatorname{ad}_X^n Y = \sum_{j=0}^n (-1)^j \binom{n}{j} X^{n-j} Y X^j. \quad (4.4)$$

Indeed, left multiplication by X commutes with right multiplication by X , so their difference satisfies the binomial theorem. This also supplies a second direct verification of [equation \(4.1\)](#).

4.2 The derivative of the exponential

For a differentiable path $X(t)$, write $H = X'(t)$. Differentiating the absolutely convergent exponential, or differentiating degree by degree formally, gives

$$(\operatorname{d exp})_X(H) = \sum_{p, q \geq 0} \frac{X^p H X^q}{(p+q+1)!} = \int_0^1 e^{(1-u)X} H e^{uX} du. \quad (4.5)$$

The integral identity follows by expanding its two exponentials and using $\int_0^1 (1-u)^p u^q du = p!q!/(p+q+1)!$. That beta integral follows by p integrations by parts, beginning with $\int_0^1 u^q du = 1/(q+1)$.

Multiplying [equation \(4.5\)](#) on the left or right and applying [equation \(4.1\)](#) yields the complete formulas

$$e^{-X} (\operatorname{d exp})_X(H) = \int_0^1 e^{-u \operatorname{ad}_X} H du = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} \operatorname{ad}_X^n H, \quad (4.6)$$

$$(\operatorname{d exp})_X(H) e^{-X} = \sum_{n \geq 0} \frac{1}{(n+1)!} \operatorname{ad}_X^n H. \quad (4.7)$$

Define the entire functions

$$f(z) = \frac{1 - e^{-z}}{z} = \sum_{n \geq 0} \frac{(-1)^n z^n}{(n+1)!}, \quad f(-z) = \frac{e^z - 1}{z}, \quad f(0) = 1. \quad (4.8)$$

Then [equations \(4.6\) and \(4.7\)](#) are $f(\text{ad}_X)H$ and $f(-\text{ad}_X)H$, respectively. The quotient is notation for an entire function; it does not require the operator ad_X to be invertible.

4.3 Bernoulli functions, coefficients, and their full expansions

The formal reciprocal of f is

$$h(z) = \frac{z}{1 - e^{-z}} = \sum_{n \geq 0} b_n^+ z^n, \quad b_n^+ = \frac{B_n^+}{n!}. \quad (4.9)$$

We use $B_1^+ = +1/2$, the convention employed in the source article. The more usual convention with $B_1^- = -1/2$ is obtained from $h(-z) = z/(e^z - 1) = \sum b_n^- z^n$, where $b_n^- = (-1)^n b_n^+$.

Proposition 4.2. *The coefficients in [equation \(4.9\)](#) are determined at every order by*

$$b_0^+ = 1, \quad b_n^+ = - \sum_{j=0}^{n-1} \frac{(-1)^{n-j}}{(n-j+1)!} b_j^+ \quad (n \geq 1). \quad (4.10)$$

They also have the nonrecursive description

$$b_1^+ = \frac{1}{2}, \quad b_{2m+1}^+ = 0 \quad (m \geq 1), \quad b_{2m}^+ = \frac{2(-1)^{m-1} \zeta(2m)}{(2\pi)^{2m}} \quad (m \geq 1). \quad (4.11)$$

The Taylor series of h has radius 2π , and

$$h(z) = \frac{z}{2} \left(1 + \coth \frac{z}{2} \right) = 1 + \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720} + \frac{z^6}{30240} - \frac{z^8}{1209600} + \cdots. \quad (4.12)$$

Every coefficient after the displayed terms is specified by [equations \(4.10\) and \(4.11\)](#).

Proof. Equating coefficients in $f(z)h(z) = 1$ gives [equation \(4.10\)](#). Elementary manipulation gives [equation \(4.12\)](#); its part after subtracting $z/2$ is even, proving the odd-coefficient assertion.

For completeness, the partial fraction expansion needed for the even coefficients can be derived from an elementary Fourier series. Initially take real $z > 0$ and periodically extend $u \mapsto \cosh(z(u - 1/2))$ from $[0, 1]$. Twice integrating its Fourier coefficients by parts gives

$$\int_0^1 \cosh(z(u - 1/2)) e^{-2\pi i k u} du = \frac{2z \sinh(z/2)}{z^2 + 4\pi^2 k^2}.$$

The coefficients are $O(k^{-2})$, so the Fourier series converges absolutely to the continuous periodic function. Evaluating at $u = 0$ and dividing by $\sinh(z/2)$ gives

$$\coth(z/2) = \frac{2}{z} + 4z \sum_{k \geq 1} \frac{1}{z^2 + 4\pi^2 k^2}.$$

Both sides are meromorphic, and the normally convergent right side extends the identity away from their poles. Consequently

$$h(z) = 1 + \frac{z}{2} + 2z^2 \sum_{k \geq 1} \frac{1}{z^2 + 4\pi^2 k^2}. \quad (4.13)$$

For $|z| < 2\pi$ each denominator can be expanded geometrically; absolute convergence justifies interchanging the sums and gives [equation \(4.11\)](#). The nearest nonremovable poles of h are $\pm 2\pi i$, proving that the Taylor radius is exactly 2π . $\square$

The other scalar function on the source page is

$$\psi(x) = \frac{x \log x}{x-1} = 1 - \sum_{n \geq 1} \frac{(1-x)^n}{n(n+1)}, \quad |1-x| < 1, \quad (4.14)$$

with $\psi(1) = 1$. To prove its expansion, put $w = 1 - x$ and use $\log(1-w) = -\sum_{j \geq 1} w^j/j$. Multiplication by $(1-w)/(-w)$ gives the stated coefficients. Near $z = 0$, with the logarithm branch satisfying $\log(e^z) = z$,

$$\psi(e^z) = h(z) = \sum_{n \geq 0} B_n^+ \frac{z^n}{n!}. \quad (4.15)$$

The equality is formal as well. It must not be interpreted as a branch-independent identity for arbitrary complex z .

4.4 The logarithmic differential equation and the Poincaré integral

Theorem 4.3 (Integral BCH formula). *Formally, and analytically in a sufficiently small neighborhood of $(0, 0)$,*

$$Z(X, Y) = X + \int_0^1 \psi(e^{\text{ad}_X} e^{t \text{ad}_Y}) Y \, dt. \quad (4.16)$$

The operator function ψ is evaluated by its series at the identity. Equivalently, for $Z(t) = \log(e^X e^{tY})$,

$$Z'(t) = h(\text{ad}_{Z(t)})Y, \quad Z(0) = X. \quad (4.17)$$

Proof. Let $g(t) = e^X e^{tY} = e^{Z(t)}$. Its left logarithmic derivative is $g^{-1}g' = Y$. Equation [equation \(4.6\)](#) gives $f(\text{ad}_Z)Z' = Y$. Its formal inverse is $h(\text{ad}_Z)$, and analytically it is invertible near $Z = 0$, proving [equation \(4.17\)](#).

By the Campbell identity,

$$e^{\text{ad}_Z} = \text{Ad}_{e^Z} = \text{Ad}_{e^X e^{tY}} = e^{\text{ad}_X} e^{t \text{ad}_Y}.$$

For the formal series, $\log(e^{\text{ad}_Z}) = \text{ad}_Z$ because ad_Z raises total degree. For small operators the same equality holds for the local analytic logarithm. Applying [equation \(4.15\)](#) therefore identifies $h(\text{ad}_Z)$ with the operator in [equation \(4.16\)](#). Integrating from zero to one proves the formula. $\square$

This argument does not require the adjoint representation to be injective: it uses a formal identity of endomorphisms, not recovery of Z from ad_Z .

4.5 A finite all-order recurrence by total degree

There is also a recursive proof of Lie-series existence that does not use the primitive-element theorem. Start with the associative logarithm of $e^{tX}e^{tY}$, and write

$$Z(t) = \sum_{n \geq 1} t^n Z_n, \quad V(t) = X + e^{t \operatorname{ad}_X} Y = \sum_{j \geq 0} t^j V_j,$$

where

$$V_0 = X + Y, \quad V_j = \frac{\operatorname{ad}_X^j Y}{j!} \quad (j \geq 1). \quad (4.18)$$

The right logarithmic derivative of $e^{tX}e^{tY}$ is $V(t)$. Equation [equation \(4.7\)](#) consequently gives

$$Z'(t) = h(-\operatorname{ad}_{Z(t)})V(t).$$

Comparing coefficients yields

$$nZ_n = \sum_{m=0}^{n-1} b_m^- \sum_{\substack{k_0 \geq 0, k_1, \dots, k_m \geq 1 \\ k_0 + k_1 + \dots + k_m = n-1}} \operatorname{ad}_{Z_{k_1}} \cdots \operatorname{ad}_{Z_{k_m}} V_{k_0}. \quad (4.19)$$

For $m = 0$ the inner expression means V_{n-1} . Every Z_{k_i} on the right has index less than n . Thus $Z_1 = X + Y$, and induction shows that every Z_n is a rational Lie polynomial. This is an independent recursive existence proof, of the same general type of conclusion as the concise existence result discussed in [3], though no use of Eichler's argument is required here. It is also a complete computational specification of every homogeneous term. For example it gives $2Z_2 = [X, Y]$ and $3Z_3 = ([X, [X, Y]] - [Y, [X, Y]])/4$.

5 Removing the $O(Y^2)$: a complete expansion in one variable

The term $O(Y^2)$ in the source article means terms containing at least two occurrences of Y , not merely terms of total degree at least two. We now determine all of those terms. Put $A = \operatorname{ad}_X$ and $B = \operatorname{ad}_Y$.

5.1 An explicit all-terms formula

Theorem 5.1 (Operator-block expansion). *The full BCH series is*

$$Z = X + Y + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k(k+1)} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0}} \frac{A^{r_1} B^{s_1} \cdots A^{r_k} B^{s_k} Y}{(1 + \sum_i s_i) \prod_i r_i! s_i!}. \quad (5.1)$$

In particular, the term with exactly $q \geq 1$ occurrences of Y is

$$H_q = \delta_{q1} Y + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k(k+1)q} \sum_{\substack{r_i + s_i > 0 \\ \sum_i s_i = q-1}} \frac{A^{r_1} B^{s_1} \cdots A^{r_k} B^{s_k} Y}{\prod_i r_i! s_i!}, \quad Z = X + \sum_{q \geq 1} H_q. \quad (5.2)$$

At fixed total degree all the sums are finite. Any summand with $s_k > 0$ vanishes.

Proof. Expand the operator in [equation \(4.16\)](#) using [equation \(4.14\)](#). Since

$$e^A e^{tB} - I = \sum_{r+s>0} \frac{t^s}{r!s!} A^r B^s,$$

its k th power is the ordered block product in [equation \(5.1\)](#), with a factor $t^{\sum_i s_i}$. The sign from $(I - e^A e^{tB})^k$ changes the coefficient to $(-1)^{k+1}/(k(k+1))$. Integrating $t^{\sum_i s_i}$ gives $1/(1 + \sum_i s_i)$. There is one final Y , so the number of Y 's is $1 + \sum_i s_i$. This proves both equations. The last assertion follows from $B^s Y = 0$ for $s > 0$. $\square$

Unlike a finite display with an unspecified remainder, [equation \(5.2\)](#) gives every term with three, four, or arbitrarily many Y 's as well.

5.2 A triangular recurrence in the number of Y 's

There is a useful alternative, especially when X is held fixed. Define $H_0 = X$ and

$$Z(X, \varepsilon Y) = \sum_{q \geq 0} \varepsilon^q H_q.$$

[Equation \(4.17\)](#) gives, for every $q \geq 1$,

$$qH_q = \sum_{m \geq 0} b_m^+ \sum_{\substack{j_1, \dots, j_m \geq 0 \\ j_1 + \dots + j_m = q-1}} \text{ad}_{H_{j_1}} \cdots \text{ad}_{H_{j_m}} Y. \quad (5.3)$$

For $m = 0$ the contribution is Y if $q = 1$ and zero otherwise. Every index j_i is less than q , so this is triangular in the Y -degree. The outer sum may contain arbitrarily many X 's, but it is finite in every fixed total degree. It follows simply by expanding $h(\text{ad}_Z)$ and equating the coefficient of ε^{q-1} .

For $q = 1$ the only indices are zero, giving

$$\begin{aligned} H_1 &= h(A)Y = \frac{A}{1 - e^{-A}} Y = \frac{A}{2} \left(1 + \coth \frac{A}{2} \right) Y, \\ &= Y + \frac{1}{2} [X, Y] + \frac{1}{12} [X, [X, Y]] - \frac{1}{720} \text{ad}_X^4 Y + \frac{1}{30240} \text{ad}_X^6 Y + \cdots \end{aligned} \quad (5.4)$$

This proves both versions of the source article's linear-in- Y formula. The quotients and the product with $\coth$ mean the regular function h ; neither expression calls for an inverse of A at zero.

For $q = 2$, [equation \(5.3\)](#) gives the full quadratic term

$$H_2 = \frac{1}{2} \sum_{m \geq 1} b_m^+ \sum_{r=0}^{m-1} A^r [H_1, A^{m-1-r} Y]. \quad (5.5)$$

No higher Y -degree has been suppressed inside this expression.

5.3 A bivariate generating function for the complete quadratic term

Theorem 5.2 (Quadratic kernel). *Let*

$$K(u, v) = \frac{h(u)}{2u} (h(u+v) - h(v)) = \sum_{p, q \geq 0} k_{pq} u^p v^q. \quad (5.6)$$

The apparent quotient by u is a regular divided difference. Then

$$H_2 = \sum_{p,q \geq 0} k_{pq} [A^p Y, A^q Y], \quad k_{pq} = \frac{1}{2} \sum_{a=0}^p b_a^+ b_{p+q+1-a}^+ \binom{p+q+1-a}{p+1-a}. \quad (5.7)$$

Thus every coefficient of the quadratic part is an explicit finite sum of Bernoulli coefficients.

Proof. On a bracket $[A^p Y, A^q Y]$, the derivation A acts as multiplication by $u + v$ in the symbolic two-variable description: this is precisely equation (1.4). The replacement of the first Y by $H_1 = h(A)Y$ acts as $h(u)$, and A^{m-1-r} on the second Y acts as v^{m-1-r} . Hence equation (5.5) is represented by

$$\frac{h(u)}{2} \sum_{m \geq 1} b_m^+ \sum_{r=0}^{m-1} (u+v)^r v^{m-1-r}.$$

The finite geometric identity

$$\sum_{r=0}^{m-1} (u+v)^r v^{m-1-r} = \frac{(u+v)^m - v^m}{u}$$

turns this into equation (5.6). Expanding $h(u)$ and the divided difference, then extracting $u^p v^q$, gives the stated finite coefficient sum. $\square$

Because brackets are antisymmetric, K can equivalently be replaced by

$$\frac{K(u, v) - K(v, u)}{2} = \frac{1}{4} \left\{ \frac{h(u)}{u} (h(u+v) - h(v)) - \frac{h(v)}{v} (h(u+v) - h(u)) \right\}.$$

For orientation, the first terms, grouped by the number of X 's, are

$$H_2 = -\frac{1}{12} [Y, AY] - \frac{1}{24} [Y, A^2 Y] - \frac{1}{180} [Y, A^3 Y] - \frac{1}{120} [AY, A^2 Y] + O_X(4), \quad (5.8)$$

where $O_X(4)$ means at least four X 's and exactly two Y 's. The kernel equation (5.6), not this short display, is the full expansion.

5.4 Analytic meaning when X is not close to zero

In a finite-dimensional Lie group, $f(\text{ad}_X)$ is invertible exactly when ad_X has no eigenvalue $2\pi i k$ with $k \in \mathbb{Z} \setminus \{0\}$. This follows because the zeros of the entire function f are exactly those numbers, and triangularization evaluates its diagonal entries on the eigenvalues. When this condition holds, the inverse function theorem gives an exponential chart centered at X , and $\log(e^X e^{\varepsilon Y})$ has a local branch equal to X at $\varepsilon = 0$. Here $h(\text{ad}_X)$ means the inverse of $f(\text{ad}_X)$, not necessarily its Taylor series at zero. In particular, one must not evaluate the infinite Bernoulli sum in equation (5.3) literally when that series diverges. The following finite-at-each-order replacement gives the actual local analytic expansion.

Put $A = \text{ad}_X$. For operators $E_1, \dots, E_m$, let $D^m h(A)[E_1, \dots, E_m]$ be the m th Fréchet derivative of the holomorphic matrix function h . It has the explicit resolvent formula

$$D^m h(A)[E_1, \dots, E_m] = \frac{1}{2\pi i} \int_{\Gamma} h(z) \sum_{\pi \in S_m} R(z) E_{\pi(1)} R(z) \cdots E_{\pi(m)} R(z) dz, \quad R(z) = (zI - A)^{-1}. \quad (5.9)$$

Here Γ is a finite union of positively oriented contours enclosing the spectrum, with interiors on which h is holomorphic; it can be chosen to avoid every pole. For real Lie groups use the complexified adjoint operator. The result preserves the real subspace because the function and the construction are invariant under complex conjugation.

The complete local coefficients at this fixed regular X satisfy $H_1 = h(A)Y$ and, for $q \geq 2$,

$$qH_q = \sum_{m=1}^{q-1} \frac{1}{m!} \sum_{\substack{j_1, \dots, j_m \geq 1 \\ j_1 + \dots + j_m = q-1}} D^m h(A) [\text{ad}_{H_{j_1}}, \dots, \text{ad}_{H_{j_m}}] Y. \quad (5.10)$$

Both sums are finite, and only earlier H_j occur. In particular, $2H_2 = Dh(A)[\text{ad}_{h(A)Y}]Y$. To prove these formulas, expand the resolvent of $A + E$ by its uniformly convergent Neumann series on Γ for sufficiently small E . Extracting the coefficient multilinear in $E_1, \dots, E_m$ yields [equation \(5.9\)](#). Substitute $E = \sum_{j \geq 1} \varepsilon^j \text{ad}_{H_j}$ in the resulting Taylor series of $h(A + E)$ and compare powers of ε in $Z' = h(\text{ad}_Z)Y$. The m th derivative carries the factor $1/m!$, giving [equation \(5.10\)](#). Analyticity of the local inverse of the exponential ensures convergence for sufficiently small ε . This separates the all-orders local expansion around a regular X from the total-degree expansion around $(0, 0)$.

6 Convergence, logarithm branches, and local Lie groups

Formal equality does not by itself authorize substituting large matrices. Conversely, failure of a Taylor series to converge does not rule out a useful analytic continuation of its sum. This distinction is central to the examples in [\[6\]](#). We establish elementary quantitative results sufficient for every convergence claim in the source article.

6.1 Two useful convergence thresholds

Theorem 6.1 (Absolute convergence and a remainder bound). *Let X, Y belong to a unital Banach algebra, and put $s = \|X\| + \|Y\|$. If $s < \log 2$, the associative expansion [equation \(3.2\)](#) is absolutely convergent, the homogeneous BCH series converges absolutely, and its sum satisfies $e^Z = e^X e^Y$. More precisely, for every $r > 0$ with $rs < \log 2$,*

$$\sum_{n \geq 1} r^n \|Z_n(X, Y)\| \leq M_s(r) := -\log(2 - e^{rs}). \quad (6.1)$$

If $1 < r < \log 2/s$, then for $Z^{[N]} = \sum_{n=1}^N Z_n$,

$$\|Z - Z^{[N]}\| \leq r^{-(N+1)} M_s(r), \quad (6.2)$$

$$\|e^{Z^{[N]}} - e^X e^Y\| \leq e^{M_s(1)} r^{-(N+1)} M_s(r). \quad (6.3)$$

The ungrouped individual nested-commutator terms in Dynkin's formula are absolutely summable under the more conservative condition

$$\|X\| + \|Y\| < \frac{\log 2}{2}. \quad (6.4)$$

Proof. Expanding into nonempty words gives

$$\|e^X e^Y - 1\| \leq e^s - 1.$$

For $s < \log 2$ this is less than one, so the logarithm series converges. If every term in its block expansion is replaced by its norm bound, its sum is at most

$$\sum_{k \geq 1} \frac{(e^s - 1)^k}{k} = -\log(2 - e^s).$$

Replacing X, Y by rX, rY proves [equation \(6.1\)](#). Absolute convergence permits grouping by total degree, and the formal exponential-logarithm identity then evaluates to $e^Z = e^X e^Y$.

The tail satisfies $\sum_{n > N} \|Z_n\| \leq r^{-(N+1)} \sum_{n > N} r^n \|Z_n\|$, which proves [equation \(6.2\)](#). The identity

$$e^A - e^B = \int_0^1 e^{(1-u)A} (A - B) e^{uB} du$$

follows by differentiating $e^{(1-u)A} e^{uB}$. Expanding the two exponentials around the middle factor gives $\|e^A - e^B\| \leq e^{\max(\|A\|, \|B\|)} \|A - B\|$. Both Z and $Z^{[N]}$ have norm at most $M_s(1)$, proving [equation \(6.3\)](#).

For a right-nested bracket of length n ,

$$\|R(a_1 \cdots a_n)\| \leq 2^{n-1} \prod_{j=1}^n \|a_j\|.$$

Thus the absolute sum of the individual Dynkin terms is bounded by

$$\frac{1}{2} \sum_{k \geq 1} \frac{(e^{2s} - 1)^k}{k} = -\frac{1}{2} \log(2 - e^{2s}),$$

where we discarded the helpful factor $1/n$. It is finite under [equation \(6.4\)](#). □

All coefficient bounds use only nonempty products and submultiplicativity. Thus the thresholds apply to any submultiplicative matrix norm, including an unnormalized Hilbert–Schmidt norm. The distinction between $\log 2$ and $(\log 2)/2$ is not a contradiction: the latter estimate controls a particular redundant commutator expansion term by term. Neither bound is asserted to be optimal.

The extra condition $\|Z\| < \log 2$ displayed next to the associative formula in the source article is a sufficient condition for taking the power-series logarithm of e^Z when Z is given in advance. It is not needed to *define* $Z = \log(e^X e^Y)$ by the convergent logarithm above. In particular, one should not silently infer an additional norm bound on this constructed Z ; [equation \(6.1\)](#) is the bound actually proved.

For an arbitrary finite-dimensional normed Lie algebra, choose $\kappa > 0$ with $\|[U, V]\| \leq \kappa \|U\| \|V\|$. The same proof replaces 2^{n-1} by κ^{n-1} , so Dynkin’s Lie series converges whenever $\kappa(\|X\| + \|Y\|) < \log 2$. This proves a small-neighborhood convergence assertion without embedding the Lie algebra into matrices.

6.2 The trace identity and its precise scope

Proposition 6.2. *For the formal BCH series, and whenever its homogeneous series converges in a matrix algebra,*

$$\operatorname{tr} Z(X, Y) = \operatorname{tr} X + \operatorname{tr} Y. \tag{6.5}$$

For an arbitrary complex matrix logarithm L satisfying $e^L = e^X e^Y$, one only has the branch-independent conclusion

$$\operatorname{tr} L - \operatorname{tr} X - \operatorname{tr} Y \in 2\pi i \mathbb{Z}. \tag{6.6}$$

Proof. The cyclic identity $\text{tr}(AB) = \text{tr}(BA)$ follows directly by interchanging the two indices in the matrix-entry sum. Every bracket therefore has trace zero. By the formal BCH theorem all Z_n , $n \geq 2$, are sums of brackets. This proves [equation \(6.5\)](#) degree by degree, and continuity of trace proves it for a convergent sum.

To prove [equation \(6.6\)](#), triangularize a complex matrix A . Its exponential has diagonal entries e^{λ_j} , where λ_j are the diagonal entries of A , so $\det(e^A) = e^{\text{tr} A}$. Determinants in $e^L = e^X e^Y$ give $e^{\text{tr} L - \text{tr} X - \text{tr} Y} = 1$, which is equivalent to [equation \(6.6\)](#). $\square$

For a scalar counterexample to unrestricted use of the principal logarithm, take $X = Y = 3\pi i/4$. Then $e^X e^Y = -i$ has principal logarithm $-\pi i/2$, whereas $X + Y = 3\pi i/2$. The difference is $-2\pi i$. When all three logarithms are real matrices, the real trace difference in [equation \(6.6\)](#) must, of course, be zero.

6.3 The article’s nonconvergence example: all logarithms classified

Theorem 6.3. *Let*

$$X = \begin{pmatrix} 0 & \pi i \\ \pi i & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (6.7)$$

Then

$$e^X e^Y = \begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix} = -I - Y. \quad (6.8)$$

All of its complex matrix logarithms are

$$L = (2k + 1)\pi i I + Y, \quad k \in \mathbb{Z}. \quad (6.9)$$

None belongs to $\mathfrak{sl}_2(\mathbb{C})$. In particular the homogeneous BCH series at this pair does not converge, even conditionally.

Proof. Since $X^2 = -\pi^2 I$, separating even and odd terms of its exponential gives $e^X = \cos \pi I + (\sin \pi / \pi) X = -I$. Since $Y^2 = 0$, $e^Y = I + Y$, proving [equation \(6.8\)](#).

Any matrix commutes with its exponential, so a logarithm L commutes with $-I - Y$ and hence with Y . Solving $LY = YL$ entry by entry gives $L = aI + bY$. Therefore $e^L = e^a(I + bY)$. Equality with $-I - Y$ forces $e^a = -1$ and $b = 1$, proving [equation \(6.9\)](#). Their traces are $2(2k + 1)\pi i$, never zero.

It remains to justify the conclusion about a possibly conditional BCH sum. Suppose $\sum_{n \geq 1} Z_n(X, Y)$ converged to L . Its coefficients would be bounded, so $\sum t^n Z_n$ would be analytic for $|t| < 1$. Near $t = 0$ its exponential is $e^{tX} e^{tY}$; the identity theorem extends that equality throughout $|t| < 1$. Abel’s limit theorem gives $\sum t^n Z_n \rightarrow L$ as real $t \uparrow 1$. For clarity, this theorem follows here from

$$\sum_{n \geq 1} t^n Z_n = (1 - t) \sum_{n \geq 1} t^n S_n, \quad S_n = \sum_{j=1}^n Z_j,$$

by splitting off a finite number of S_n and bounding the tail of $S_n - L$. Continuity of the exponential would give $e^L = -I - Y$. But all Z_n are traceless, so their limit is traceless, contradicting the classification of logarithms. $\square$

The obstruction is not that this invertible complex matrix has no logarithm. It has infinitely many. The obstruction is that it has no logarithm in the specified traceless Lie algebra.

6.4 Exponential coordinates in an arbitrary Lie group

For a finite-dimensional Lie group G , identify its Lie algebra with the tangent space at the identity and define the bracket by left-invariant vector fields. The adjoint action is the differential of conjugation. Differentiating the one-parameter group of conjugations gives

$$\frac{d}{dt} \text{Ad}_{\exp(tX)} = \text{ad}_X \text{Ad}_{\exp(tX)}, \quad \text{Ad}_{\exp(0X)} = I,$$

so $\text{Ad}_{\exp X} = e^{\text{ad}_X}$ intrinsically as well.

The differential of the exponential has the same form as [equation \(4.6\)](#). One intrinsic derivation is useful. Let $F(s, t) = \exp(s(X + tH))$, let θ be the left Maurer–Cartan form, and put $U(s) = \theta(\partial_t F(s, 0))$. The velocity in the s direction is $X + tH$. The Maurer–Cartan identity, proved in [section 10](#), gives

$$U'(s) = H - [X, U(s)], \quad U(0) = 0.$$

Its solution is $U(1) = \int_0^1 e^{-u \text{ad}_X} H \, du = f(\text{ad}_X)H$. In particular $(d \exp)_0 = I$. The inverse function theorem provides neighborhoods $U \subset \mathfrak{g}$ of zero and $V \subset G$ of the identity on which $\exp : U \rightarrow V$ is a diffeomorphism.

For sufficiently small X, Y , the curve $\log(\exp X \exp(tY))$ in this chart satisfies exactly [equation \(4.17\)](#). The convergent universal Lie series also satisfies that equation, by termwise differentiation on a smaller neighborhood. Uniqueness of the ordinary differential equation identifies them. Thus

$$\exp X \exp Y = \exp(\text{BCH}(X, Y)) \tag{6.10}$$

in every finite-dimensional Lie group for sufficiently small X, Y . This proves both local exponential parametrization and determination of the local multiplication by the Lie bracket.

Proposition 6.4 (Local and simply connected integration). *A Lie algebra homomorphism $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ induces the local group homomorphism*

$$\exp_G X \mapsto \exp_H(\phi X). \tag{6.11}$$

If G is connected and simply connected, this local homomorphism extends uniquely to a smooth homomorphism $G \rightarrow H$ with differential ϕ .

Proof. Since ϕ preserves every iterated bracket, it preserves every Z_n , and hence the convergent BCH sum. Equation [equation \(6.10\)](#) therefore proves local multiplicativity of [equation \(6.11\)](#) whenever the factors and their product lie in a sufficiently small exponential neighborhood.

Here is the extension argument. For a path γ from the identity to g , partition its parameter interval so finely that every increment $\gamma(t_{j-1})^{-1}\gamma(t_j)$ lies in that neighborhood. Define the candidate image of g as the ordered product of the local images of these increments. Refining a sufficiently fine partition does not change the product, by local multiplicativity. In a homotopy of paths, choose a rectangular grid fine enough that all increments and all products within each grid cell stay in the same neighborhood. The two routes across a cell have the same local image, because the corresponding group increments have the same product. Cancelling internal edges shows that the products for the two boundary paths agree. Simple connectedness therefore makes the result independent of the path.

Concatenating a path to g with the left translate by g of a path to h proves multiplicativity. Near any point the construction is a translate of the original smooth local map, so it is smooth with differential ϕ . Uniqueness holds because a homomorphism with that differential agrees on one-parameter subgroups, hence on a neighborhood of the identity; a connected group is generated by every identity neighborhood. $\square$

This gives the representation-theoretic implication mentioned in the source article by taking $H = \text{GL}(V)$. Simple connectedness matters for the global extension: local Lie-algebra information alone does not determine a general group's topology. For example $(\mathbb{R}, +)$ and the circle group have the same one-dimensional abelian Lie algebra but are not isomorphic, since the latter is compact and the former is not.

6.5 Nilpotence gives an exact polynomial group law

If a Lie algebra has nilpotency class at most c , all brackets of length greater than c vanish, so $Z_n = 0$ for $n > c$. The finite polynomial $X * Y = \text{BCH}(X, Y)$ then defines a group law on its underlying vector space. To verify associativity, use formal exponentials in three free variables:

$$e^{\text{BCH}(\text{BCH}(X,Y),Z)} = e^X e^Y e^Z = e^{\text{BCH}(X,\text{BCH}(Y,Z))}.$$

Formal logarithms give the identity, which evaluates as a finite identity in the nilpotent algebra. Likewise 0 is the identity and $-X$ is the inverse of X . Smoothness is polynomial, the vector space is simply connected, and the leading term of the group commutator is $ts[X, Y]$, so its Lie algebra is the original one. Since $(tX) * (uX) = (t + u)X$, the exponential map in this group is the identity in these coordinates. Applying [theorem 6.4](#) in both directions shows that any simply connected Lie group with this nilpotent Lie algebra is isomorphic to this polynomial group. In particular its exponential map is a global diffeomorphism.

7 Exactly summable commutator relations

The universal series is infinite, but algebraic relations can remove almost all of it. The following proofs establish global exponential identities, not just small-neighborhood Taylor expansions. For Banach algebras all exponentials exist; for finite-dimensional Lie groups the same proofs use one-parameter subgroups and their logarithmic derivatives. Complex scalar parameters are understood in a complex Banach algebra or complex Lie group; in a real Lie group the parameters and linear combinations must be real.

7.1 Commuting elements and central commutators

If $[X, Y] = 0$, the binomial proof in [theorem 1.1](#) gives

$$e^X e^Y = e^{X+Y} \tag{7.1}$$

without a smallness restriction.

Theorem 7.1 (Central-commutator or disentangling identity). *Suppose $C = [X, Y]$ commutes with both X and Y . Then, for every scalar parameter t for which the exponentials are interpreted,*

$$e^{tX} Y e^{-tX} = Y + tC, \tag{7.2}$$

$$e^{tX} e^{tY} = \exp\left(t(X + Y) + \frac{1}{2}t^2 C\right). \tag{7.3}$$

In particular,

$$e^X e^Y = e^{X+Y+C/2}, \quad e^{X+Y} = e^X e^Y e^{-C/2}, \quad e^{X/2} e^Y e^{X/2} = e^{X+Y}. \tag{7.4}$$

Proof. All terms of [equation \(4.1\)](#) after the first commutator vanish, proving [equation \(7.2\)](#). If $g(t) = e^{tX} e^{tY}$, then

$$g'(t) = (X + e^{tX} Y e^{-tX})g(t) = (X + Y + tC)g(t), \quad g(0) = I. \tag{7.5}$$

The exponent $K(t) = t(X + Y) + t^2 C/2$ commutes with its derivative, so differentiating its exponential gives the same equation and initial condition. Uniqueness proves [equation \(7.3\)](#). Taking $t = 1$ gives the first identity in [equation \(7.4\)](#), and commuting the central exponential gives the second. Applying the first identity twice with $X/2, Y, X/2$ cancels the two central corrections, proving the symmetric identity. $\square$

Centrality is needed only relative to the Lie subalgebra generated by X, Y , not relative to a larger ambient algebra.

A useful complete finite-product extension is the following. If all commutators $[X_i, X_j]$ commute with every X_k , then

$$\prod_{j=1}^m e^{X_j} = \exp\left(\sum_{j=1}^m X_j + \frac{1}{2} \sum_{1 \leq i < j \leq m} [X_i, X_j]\right). \quad (7.6)$$

The proof is induction on m using [theorem 7.1](#); the new commutator is the sum of the pairwise commutators with the last element. This is the all-factors form behind the Weyl and displacement multiplication laws.

7.2 An eigenvector of the adjoint action

Theorem 7.2 (The relation $[X, Y] = sY$). *Suppose $[X, Y] = sY$ with $s \in \mathbb{C}$. Define*

$$q_s(t) = \begin{cases} (1 - e^{-st})/s, & s \neq 0, \\ t, & s = 0. \end{cases}$$

Then for every scalar c ,

$$e^{t(X+cY)} = e^{tX} e^{cq_s(t)Y}. \quad (7.7)$$

Consequently, when $s \notin 2\pi i\mathbb{Z} \setminus \{0\}$,

$$\boxed{e^X e^Y = \exp\left(X + \frac{s}{1 - e^{-s}} Y\right)}, \quad (7.8)$$

with the coefficient interpreted as 1 at $s = 0$. For every s , including resonant values,

$$e^X e^Y = e^{e^s Y} e^X, \quad e^X e^Y e^{-X} = e^{e^s Y}. \quad (7.9)$$

Proof. Equation [equation \(4.1\)](#) gives $e^{tX} Y e^{-tX} = e^{st} Y$. The right logarithmic derivative of $F(t) = e^{tX} e^{cq_s(t)Y}$ is therefore

$$F' F^{-1} = X + cq'_s(t) e^{st} Y = X + cY,$$

because $q'_s(t) = e^{-st}$. This proves [equation \(7.7\)](#) by uniqueness of the differential equation with $F(0) = I$. At $t = 1$, choose $c = 1/q_s(1) = s/(1 - e^{-s})$ when this is defined. The apparent singularity at zero is removable, and the nonzero zeros of $1 - e^{-s}$ are exactly the excluded values. The braiding formulas follow directly from conjugating the exponential and require no division by $q_s(1)$. $\square$

Every bracket containing at least two Y 's vanishes under this relation. To see this, evaluate any bracket tree. A branch containing one or more Y 's, when nonzero, is a multiple of Y . At the first node combining two such branches the bracket is zero. A branch consisting only of X 's is zero unless it has length one. Thus only the terms linear in Y survive, and formally

$$Z = X + \sum_{n \geq 0} b_n^+ s^n Y. \quad (7.10)$$

This also follows from [equation \(5.4\)](#), but the differential-equation proof establishes [equation \(7.8\)](#) even when that Taylor series diverges.

If $Y \neq 0$, [equation \(4.11\)](#) shows that [equation \(7.10\)](#) converges for $|s| < 2\pi$ and diverges for $|s| \geq 2\pi$: its even-index terms have magnitudes proportional to $2\zeta(2m)(|s|/(2\pi))^{2m}$ and therefore do not tend to zero in the latter range. The closed expression [equation \(7.8\)](#) can still be valid there.

For an explicit illustration take

$$X = \begin{pmatrix} s & 0 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (7.11)$$

Direct exponentiation gives

$$e^X e^Y = \begin{pmatrix} e^s & e^s \\ 0 & 1 \end{pmatrix}, \quad e^{X+cY} = \begin{pmatrix} e^s & c(e^s - 1)/s \\ 0 & 1 \end{pmatrix},$$

with the usual continuous value at $s = 0$. If $s = 3\pi$, the choice $c = h(s)$ works although the BCH series diverges. If $s = 2\pi i k \neq 0$, every e^{X+cY} is I , whereas $e^X e^Y = I + Y$. This shows what fails at resonance: not the braiding identity, but the possibility of a logarithm of the particular form $X + cY$. In this example a different logarithm, namely Y , still exists.

8 The Zassenhaus formula to every order

BCH replaces a product of exponentials by one exponential. Zassenhaus factorization performs an ordered operation in the opposite direction. We derive both its existence and a finite all-order Lie recurrence, then prove a quantitative convergence condition. The logarithmic-derivative approach is closely related to the recursive constructions in [\[5\]](#); the derivation here includes all steps.

8.1 Existence, uniqueness, and an explicit coefficient recursion

Theorem 8.1 (Formal Zassenhaus factorization). *There exist unique homogeneous Lie polynomials $C_n(X, Y)$ of degree n , $n \geq 2$, such that*

$$e^{t(X+Y)} = e^{tX} e^{tY} \prod_{n=2}^{\infty} e^{t^n C_n(X, Y)}. \quad (8.1)$$

The product is ordered by increasing n from left to right and is well defined formally. If

$$R_1(t) = e^{-tY} e^{-tX} e^{t(X+Y)}, \quad (8.2)$$

then every exponent is determined by

$$C_n = [t^n] R_{n-1}(t) = [t^n] \log R_{n-1}(t), \quad R_n(t) = e^{-t^n C_n} R_{n-1}(t), \quad n \geq 2. \quad (8.3)$$

Proof. The linear term of R_1 is zero, so $R_1 = 1 + O(t^2)$. Suppose inductively $R_{n-1} = 1 + t^n C_n + O(t^{n+1})$. The leading coefficient of its logarithm equals C_n , and multiplication on the left by $e^{-t^n C_n}$ removes this coefficient. Hence $R_n = 1 + O(t^{n+1})$.

Each residual is group-like in the completed free associative Hopf algebra: this holds initially, and remains true after multiplication by the exponential of a Lie polynomial. Its logarithm is primitive, so the leading homogeneous coefficient C_n is a Lie polynomial by [theorem 2.3](#). This proves the induction as well as the factorization after any fixed truncation. Uniqueness follows because the first degree at which two ordered factorizations could differ is exactly the degree of the first differing exponent. $\square$

For a direct finite formula, extracting t^n in [equation \(8.3\)](#) gives

$$C_n = \sum_{\substack{k_2, \dots, k_{n-1}, a, b, c \geq 0 \\ \sum_{j=2}^{n-1} j k_j + a + b + c = n}} \left(\prod_{j=n-1}^2 \frac{(-C_j)^{k_j}}{k_j!} \right) \frac{(-Y)^a}{a!} \frac{(-X)^b}{b!} \frac{(X+Y)^c}{c!}. \quad (8.4)$$

The descending product order is essential. This is a finite recurrence in previously computed homogeneous polynomials. It involves associative products, but the result is a Lie polynomial by the theorem. Together with the projection [equation \(2.4\)](#), it is a complete nested-commutator prescription.

The compact variant on the source page is recovered by setting $C_1 = Y$:

$$e^{-tX} e^{t(X+Y)} = \prod_{n \geq 1} e^{t^n C_n}.$$

There is no missing e^{tY} when this indexing convention is used.

8.2 A recurrence consisting entirely of commutators

Define the right logarithmic derivatives

$$F_n(t) = R'_n(t) R_n(t)^{-1}. \quad (8.5)$$

Differentiating [equation \(8.2\)](#), and using conjugation, gives

$$\begin{aligned} F_1(t) &= -Y - e^{-t \operatorname{ad}_Y} X + e^{-t \operatorname{ad}_Y} e^{-t \operatorname{ad}_X} (X + Y) \\ &= e^{-t \operatorname{ad}_Y} (e^{-t \operatorname{ad}_X} - I) Y = \sum_{k \geq 1} t^k f_{1,k}, \end{aligned} \quad (8.6)$$

where the complete initial coefficients are

$$f_{1,k} = (-1)^k \sum_{j=1}^k \frac{\operatorname{ad}_Y^{k-j} \operatorname{ad}_X^j Y}{(k-j)! j!}. \quad (8.7)$$

Since $R_{n-1} = 1 + t^n C_n + O(t^{n+1})$, one has $F_{n-1} = n t^{n-1} C_n + O(t^n)$. Differentiating the residual update yields

$$\boxed{C_n = \frac{1}{n} [t^{n-1}] F_{n-1}, \quad F_n = e^{-t^n \operatorname{ad}_{C_n}} (F_{n-1} - n t^{n-1} C_n).} \quad (8.8)$$

Indeed, the derivative of the new left factor contributes $-n t^{n-1} C_n$, while conjugating the old logarithmic derivative contributes $e^{-t^n \operatorname{ad}_{C_n}} F_{n-1}$. The correction commutes with C_n , so it may be moved inside this operator exponential.

Writing $F_n = \sum_{k \geq 0} t^k f_{n,k}$, with nonexistent coefficients defined as zero, gives a fully finite coefficient recurrence:

$$f_{n,k} = \sum_{j=0}^{\lfloor k/n \rfloor} \frac{(-1)^j}{j!} \operatorname{ad}_{C_n}^j f_{n-1, k-nj} - n C_n \delta_{k, n-1}, \quad C_n = \frac{f_{n-1, n-1}}{n}. \quad (8.9)$$

These identities prove and determine every Zassenhaus term directly in the Lie algebra. No arbitrary-order ellipsis remains to be guessed.

8.3 The exponents displayed on the source page

From equations (8.7) and (8.8),

$$C_2 = -\frac{1}{2}[X, Y], \quad (8.10)$$

$$C_3 = \frac{1}{6}[X, [X, Y]] + \frac{1}{3}[Y, [X, Y]], \quad (8.11)$$

$$C_4 = -\frac{1}{24}[X, [X, [X, Y]]] - \frac{1}{8}[Y, [X, [X, Y]]] - \frac{1}{8}[Y, [Y, [X, Y]]]. \quad (8.12)$$

For C_2 and C_3 the result is immediate from $f_{1,1}/2$ and $f_{1,2}/3$. For C_4 , the potential update at cubic order is proportional to $[C_2, f_{1,1}] = [C_2, 2C_2] = 0$, so $C_4 = f_{1,3}/4$. This proves all three formulas, including their coefficients.

Twice applying antisymmetry rewrites equation (8.12) as

$$C_4 = -\frac{1}{24}([[[X, Y], X], X] + 3[[[X, Y], X], Y] + 3[[[X, Y], Y], Y]), \quad (8.13)$$

exactly the left-nested version in the source. Substituting equations (8.10), (8.11) and (8.13) into equation (8.1) gives all of its displayed factors. Higher exponents are given by equation (8.9); additional explicit fifth- and sixth-degree expressions are included in section B, and all coefficients through degree twelve accompany the paper.

There is a left-oriented companion:

$$e^{t(X+Y)} = \dots e^{t^4 \widehat{C}_4} e^{t^3 \widehat{C}_3} e^{t^2 \widehat{C}_2} e^{tY} e^{tX}, \quad \widehat{C}_n = (-1)^{n+1} C_n. \quad (8.14)$$

It follows by replacing t with $-t$ in equation (8.1) and taking inverses, which reverses the factor order.

8.4 A self-contained convergence majorant

Formal stabilization of a product does not automatically imply analytic convergence. The following bound supplies a concrete domain; it is deliberately elementary rather than optimal.

Theorem 8.2 (A convergent Zassenhaus product). *Let $s = \|X\| + \|Y\|$. If*

$$e^{2s|t|} - 1 - 2s|t| < 2 \log 2 - 1, \quad (8.15)$$

then $\sum_{n \geq 2} |t|^n \|C_n\| < \infty$, the ordered product in equation (8.1) converges, and the identity holds in the Banach algebra. In particular the condition is satisfied when

$$|t|s < 0.3834990014 \dots,$$

where twice this constant is the positive solution u of $e^u - 1 - u = 2 \log 2 - 1$.

Proof. Write $R_1(t) = 1 + \sum_{n \geq 2} t^n P_n$. Expanding its three factors gives

$$\|P_n\| \leq \frac{(2s)^n}{n!}, \quad n \geq 2.$$

Thus $p(z) = e^{2sz} - 1 - 2sz$ is a nonnegative coefficient majorant for $R_1 - 1$.

Let $a(z) = \sum_{n \geq 2} a_n z^n$ be the unique formal nonnegative-coefficient solution of

$$a = p + (e^a - 1 - a), \quad \text{equivalently} \quad 2a - e^a + 1 = p. \quad (8.16)$$

Its coefficients are determined successively because the nonlinear expression $e^a - 1 - a$ has at least two factors of a , each of order at least two. Comparing the coefficient of t^n in $R_1 = \prod_{j \geq 2} e^{t^j C_j}$ gives C_n equal to P_n minus terms using only C_j with $j < n$. The sum of norms of those terms is bounded by the corresponding coefficient of $e^a - 1 - a$. Induction proves $\|C_n\| \leq a_n$.

For a real $r \geq 0$ satisfying [equation \(8.15\)](#), there is a unique $a_* \in [0, \log 2)$ with $2a_* - e^{a_*} + 1 = p(r)$, because the left side increases from 0 to $2 \log 2 - 1$ on that interval. Iterate $a^{(0)} = 0$ and $a^{(m+1)} = p + e^{a^{(m)}} - 1 - a^{(m)}$. The formal coefficients increase and stabilize degree by degree at those of a . Their numerical values at r increase and remain bounded by a_* . Hence, by monotone convergence of nonnegative sums, $\sum a_n r^n \leq a_* < \infty$. It follows that $\sum r^n \|C_n\| < \infty$.

Products of $e^{t^n C_n}$ consequently converge: expanding the factors bounds all products of nonconstant terms by $\exp(\sum |t|^n \|C_n\|)$, and the tail tends to one. Absolute convergence permits collecting coefficients, which agree formally with $e^{t(X+Y)}$ by [theorem 8.1](#). This proves the analytic identity. $\square$

8.5 Entire resummation in the eigenvector case

When $[X, Y] = sY$, [equation \(7.7\)](#) gives

$$R_1(t) = e^{-tY} e^{q_s(t)Y} = e^{(q_s(t)-t)Y}.$$

All factors are multiples of the same Y and commute. Uniqueness in [theorem 8.1](#) therefore yields the all-order closed expression

$$C_n = \frac{(-1)^{n-1} s^{n-1}}{n!} Y, \quad n \geq 2. \quad (8.17)$$

Its sum and product converge for every finite s, t . This is a useful contrast with the Bernoulli BCH series [equation \(7.10\)](#), whose radius in s is only 2π . If the commutator is central, [equation \(7.4\)](#) instead gives $C_2 = -[X, Y]/2$ and $C_n = 0$ for every $n \geq 3$.

9 Trotter and symmetric product formulas

The source page mentions the Suzuki–Trotter decomposition without displaying a particular version. We prove the basic product limit, the symmetric second-order formula, and a recursive construction of arbitrarily high even order. All analytic statements in this section concern matrices or bounded elements of a Banach algebra.

9.1 The product limit and its complete effective exponent

Theorem 9.1 (Lie–Trotter formula). *For every fixed t ,*

$$e^{t(X+Y)} = \lim_{n \rightarrow \infty} (e^{tX/n} e^{tY/n})^n. \quad (9.1)$$

For sufficiently large n the following is an exact convergent expansion:

$$(e^{tX/n} e^{tY/n})^n = \exp \left(t(X+Y) + \sum_{k \geq 2} \frac{t^k}{n^{k-1}} Z_k(X, Y) \right). \quad (9.2)$$

The error in [equation \(9.1\)](#) is $O(n^{-1})$, uniformly for t in any fixed compact set.

Proof. For large n , the step size is inside the convergence domain of [theorem 6.1](#). Homogeneity gives $\log(e^{tX/n}e^{tY/n}) = \sum_{k \geq 1} (t/n)^k Z_k$. Taking the n th power multiplies this one logarithm by n , since its exponential commutes with itself. This proves [equation \(9.2\)](#). Its exponent differs from $t(X + Y)$ by $O(n^{-1})$, uniformly on compact t sets by the majorant. The exponential difference estimate in the proof of [theorem 6.1](#) proves the limit and its error bound. $\square$

Thus the Zassenhaus viewpoint explains which correction factors are discarded, while BCH provides a particularly direct proof of the limit and all corrections to its effective exponent.

9.2 Symmetry removes every even degree

Put

$$S_2(t) = e^{tX/2} e^{tY} e^{tX/2}. \quad (9.3)$$

Because $S_2(-t) = S_2(t)^{-1}$, its formal or local logarithm is odd:

$$\log S_2(t) = t(X + Y) + \sum_{j \geq 1} t^{2j+1} E_{2j+1}. \quad (9.4)$$

Substituting [equations \(3.11\) to \(3.13\)](#) twice into $\text{BCH}(\text{BCH}(tX/2, tY), tX/2)$ gives

$$E_3 = -\frac{1}{24}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]]. \quad (9.5)$$

For an explicit coefficient check, first put $A = tX/2$, $B = tY$ and expand $\text{BCH}(A, B) = A + B + [A, B]/2 + [A, [A, B]]/12 + [B, [B, A]]/12 + O(t^4)$. A second BCH with A adds $[B, A]/2$ at degree two, cancelling the first correction. The degree-three terms collect to $-[A, [A, B]]/6 + [B, [B, A]]/6$; replacing A, B by $tX/2, tY$ gives [equation \(9.5\)](#).

Exactly as for Trotter,

$$S_2(t/n)^n = \exp\left(t(X + Y) + \sum_{j \geq 1} \frac{t^{2j+1}}{n^{2j}} E_{2j+1}\right) \longrightarrow e^{t(X+Y)}, \quad (9.6)$$

with error $O(n^{-2})$. Every E_{2j+1} is determined by iterated BCH, or by the following full multi-exponential formula.

For arbitrary $A_1, \dots, A_m$, expansion of the ordinary logarithm gives

$$\log\left(\prod_{i=1}^m e^{tA_i}\right) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_{ji} \geq 0 \\ \sum_i r_{ji} > 0 \ (1 \leq j \leq k)}} \frac{t^{\sum_{j,i} r_{ji}}}{\prod_{j,i} r_{ji}!} \prod_{j=1}^k \left(\prod_{i=1}^m A_i^{r_{ji}}\right). \quad (9.7)$$

It follows by writing the product minus one as its nonconstant exponential-block series and taking its logarithm, exactly as in [theorem 3.1](#). Coefficient extraction in [equation \(9.7\)](#) is finite at every order, and the primitive-element proof applies to any finite alphabet, so every homogeneous result is a Lie polynomial. Taking $(A_1, A_2, A_3) = (X/2, Y, X/2)$ determines every term of [equation \(9.4\)](#) without an unspecified remainder.

9.3 Arbitrarily high even order by a five-factor recursion

The recursive symmetric construction below is the five-factor Suzuki scheme [8]; its order conditions follow directly from BCH.

Theorem 9.2 (Symmetric recursive splitting). *Suppose a symmetric product S_{2k} satisfies*

$$S_{2k}(-t) = S_{2k}(t)^{-1}, \quad \log S_{2k}(t) = tH + t^{2k+1}E + O(t^{2k+3}), \quad H = X + Y.$$

Define

$$p_k = \frac{1}{4 - 4^{1/(2k+1)}}, \quad (9.8)$$

$$S_{2k+2}(t) = S_{2k}(p_k t)^2 S_{2k}((1 - 4p_k)t) S_{2k}(p_k t)^2. \quad (9.9)$$

Then S_{2k+2} is symmetric and $\log S_{2k+2}(t) = tH + O(t^{2k+3})$. Starting from equation (9.3), this constructs every even order, and $S_{2k}(t/n)^n = e^{tH} + O(n^{-2k})$ for fixed t .

Proof. The palindromic order of the factors proves symmetry. Up to degree $2k + 1$, their logarithms add: commutators of their leading terms vanish since all are multiples of H , and a commutator involving one error term has degree at least $2k + 2$. The coefficient of tH is $4p_k + (1 - 4p_k) = 1$. The coefficient of $t^{2k+1}E$ is

$$4p_k^{2k+1} + (1 - 4p_k)^{2k+1} = 0,$$

by equation (9.8). Symmetry makes the logarithm odd, so there is no degree $2k + 2$ term either. This gives the claimed order. The global n -step estimate follows by multiplying the local logarithm by n , as in theorem 9.1. $\square$

The middle time coefficient is negative; that poses no problem for bounded exponentials, but is a reason not to transplant this statement without additional hypotheses to arbitrary one-sided semigroups. Every higher effective-exponent coefficient of these products is explicitly obtainable from equation (9.7). If $[X, Y]$ is central, equation (7.4) shows that already $S_2(t) = e^{t(X+Y)}$ exactly.

10 Exponential coordinates, coframes, and metrics to all orders

The differential identity has an intrinsic geometric meaning. This section also isolates the corrections needed in the geometric paragraph of the source article.

10.1 The Maurer–Cartan form and its complete coordinate expansion

Let G be a finite-dimensional Lie group, with basis $e_1, \dots, e_d$ of $\mathfrak{g}$ and structure constants

$$[e_i, e_j] = f_{ij}^k e_k. \quad (10.1)$$

Repeated upper–lower indices are summed. The left Maurer–Cartan form is the $\mathfrak{g}$ -valued one-form $\theta_g = (L_{g^{-1}})_*$; in a matrix group it is $g^{-1}dg$. For $g = \exp X$, where $X = X^i e_i$, equation (4.6) gives

$$\theta = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)!} \text{ad}_X^n(dX) = e_a W^a_j(X) dX^j, \quad (10.2)$$

$$M^a_j = X^i f_{ij}^a, \quad W = f(M) = \sum_{n=0}^{\infty} \frac{(-1)^n M^n}{(n+1)!}. \quad (10.3)$$

The first terms are, with all contractions visible,

$$\theta = dX^i e_i - \frac{1}{2} X^i dX^j f_{ij}^k e_k + \frac{1}{6} X^i X^j dX^k f_{jk}^\ell f_{i\ell}^m e_m - \dots \quad (10.4)$$

For arbitrary $n \geq 1$ the contraction in the n th term is specified by

$$(M^n)^a_j = X^{i_1} \dots X^{i_n} f_{i_n j}^{b_1} f_{i_{n-1} b_1}^{b_2} \dots f_{i_1 b_{n-1}}^a, \quad (10.5)$$

with the evident interpretation when $n = 1$. Thus [equation \(10.2\)](#) is an all-terms formula, not merely an instruction to continue a pattern.

The notation

$$W = \frac{I - e^{-M}}{M} \quad (10.6)$$

means the entire function $f(M)$, not multiplication by an existing inverse of M . In fact

$$MX = \text{ad}_X X = 0, \quad M(0) = 0, \quad W(0) = I. \quad (10.7)$$

In positive dimension M is therefore always singular. The matrix supplying the coframe in exponential coordinates is W , not M . Near the identity its dual frame has coefficient matrix $W^{-1} = h(M)$. More generally,

$$W \text{ is invertible} \iff \text{spec}(M) \cap \{2\pi i k : k \in \mathbb{Z} \setminus \{0\}\} = \emptyset. \quad (10.8)$$

Indeed, complex triangularization makes the eigenvalues of $f(M)$ equal to $f(\lambda)$, counted with algebraic multiplicity. The zeros of f are precisely the indicated nonzero points. This argument also explains why the zero eigenvalue forced by [equation \(10.7\)](#) is harmless for W .

Proposition 10.1 (Maurer–Cartan equation). *The form θ satisfies*

$$d\theta + \frac{1}{2} [\theta, \theta] = 0. \quad (10.9)$$

Here $\frac{1}{2} [\theta, \theta](U, V) = [\theta(U), \theta(V)]$.

Proof. On left-invariant vector fields U, V , both $\theta(U)$ and $\theta(V)$ are constant, so the definition of exterior differentiation gives $d\theta(U, V) = -\theta([U, V]) = -[\theta(U), \theta(V)]$. Left-invariant vectors span each tangent space, proving the identity. In a matrix realization the same proof is the calculation $d(g^{-1}) = -g^{-1}(dg)g^{-1}$ and $d(g^{-1}dg) = -(g^{-1}dg) \wedge (g^{-1}dg)$. $\square$

In particular, for a two-parameter map $g(s, t)$, writing $A = \theta(\partial_s g)$ and $U = \theta(\partial_t g)$ gives $\partial_s U - \partial_t A + [A, U] = 0$. This is the mixed-variation identity used in [section 6.4](#); it involves no matrix embedding assumption.

10.2 Pullback tensors: the necessary Jacobian and nondegeneracy condition

Let B be a symmetric bilinear form on $\mathfrak{g}$. Left translation defines a symmetric tensor on G by $\langle u, v \rangle_g = B(\theta_g u, \theta_g v)$. It is a pseudo-Riemannian metric exactly when B is nondegenerate. Given a smooth map $\varphi : N \rightarrow G$, use coordinates q^α and write locally $\varphi(q) = \exp(X(q))$. Set

$$E^a_\alpha = W^a_j(X(q)) \partial_\alpha X^j. \quad (10.10)$$

The pullback tensor is

$$g_{\alpha\beta}(q) = B_{ab} E^a_\alpha E^b_\beta. \quad (10.11)$$

This is immediate by applying the definition of the pullback to the coordinate vectors. For an arbitrary, rather than left-invariant, metric on G , its coefficients B_{ab} in the left-invariant frame depend on $\varphi(q)$; the same formula holds with this dependence retained. Only when the coordinates themselves are $q^i = X^i$ does the Jacobian disappear and the expression reduce to $g_{ij} = B_{ab} W^a_i W^b_j$.

Three distinct qualifications are important. An arbitrary pullback need not be nondegenerate: a constant map gives the zero tensor. If B is positive definite, an immersion has a positive-definite pullback; for indefinite B , even an immersion can have a degenerate pullback. For example, the immersion $t \mapsto (t, t)$ into $(\mathbb{R}^2, dx^2 - dy^2)$ pulls the metric back to zero. Finally, the Killing form

$$B_K(U, V) = \text{tr}(\text{ad}_U \text{ad}_V) \quad (10.12)$$

need not be nondegenerate. In an abelian Lie algebra it vanishes identically. Consequently, choosing it does not automatically give a metric or a vielbein construction.

For completeness, the Killing form is symmetric by cyclicity of trace, and it is infinitesimally invariant:

$$B_K([X, U], V) + B_K(U, [X, V]) = 0. \quad (10.13)$$

To verify this, use $\text{ad}_{[X, U]} = [\text{ad}_X, \text{ad}_U]$ and combine the four trace terms into the trace of a commutator. The latter identity for adjoint maps is the Jacobi identity. This proves the invariance used next without assuming nondegeneracy.

10.3 All-order metric and volume formulas

Put $U_\alpha = \partial_\alpha X$. Substituting equation (10.3) in equation (10.11) gives the complete expansion

$$g_{\alpha\beta} = \sum_{r,s \geq 0} \frac{(-1)^{r+s}}{(r+1)!(s+1)!} B(M^r U_\alpha, M^s U_\beta). \quad (10.14)$$

In finite dimension this double series converges absolutely for every M and all vectors, regardless of whether the tensor it defines is nondegenerate.

Proposition 10.2 (Invariant-form simplification). *Suppose $B([X, U], V) = -B(U, [X, V])$. Then*

$$g_{\alpha\beta} = B\left(U_\alpha, \frac{2(\cosh M - I)}{M^2} U_\beta\right) \quad (10.15)$$

$$= \sum_{m=0}^{\infty} \frac{2}{(2m+2)!} B(U_\alpha, M^{2m} U_\beta). \quad (10.16)$$

The quotient denotes its entire extension. The first coefficients are 1, 1/12, 1/360, 1/20160, and so on.

Proof. Move each M in the first argument across B , picking up a minus sign. Thus $B(f(M)U, f(M)V) = B(U, f(-M)f(M)V)$. The scalar identity $f(-z)f(z) = (e^z + e^{-z} - 2)/z^2$ then proves both formulas by power-series multiplication. $\square$

Notice that all odd orders disappear only under the stated invariance condition; no such cancellation was assumed in equation (10.14).

When B is nondegenerate and $X^1, \dots, X^d$ are valid exponential coordinates, the metric matrix is $W^T B W$. Its volume density is consequently

$$\sqrt{|\det g|} = |\det W| \sqrt{|\det B|}, \quad \det W = \prod_{\lambda \in \text{spec } M} \frac{1 - e^{-\lambda}}{\lambda}, \quad (10.17)$$

where the factor at zero is 1 and multiplicities are included. Near $M = 0$ its logarithm has a second complete expansion:

$$\log \det W = -\frac{1}{2} \operatorname{tr} M + \sum_{m=1}^{\infty} \frac{b_{2m}^+}{2m} \operatorname{tr}(M^{2m}). \quad (10.18)$$

For instance the first even terms are $(\operatorname{tr} M^2)/24 - (\operatorname{tr} M^4)/2880$. To prove this, differentiate the scalar logarithm:

$$\frac{d}{dz} \log f(z) = -\frac{1}{2} + \sum_{m \geq 1} b_{2m}^+ z^{2m-1}. \quad (10.19)$$

Integration and $f(0) = 1$ give its Taylor series; triangularization and summing over eigenvalues yield [equation \(10.18\)](#). For example, $\|M\| < 2\pi$ is a sufficient condition for this Taylor series. The determinant formula [equation \(10.17\)](#) itself has no Taylor-radius restriction.

11 Weyl operators and displacement operators with domains under control

Central commutators make the algebra short, but they do not remove the domain questions for unbounded operators. We prove the identities in two explicit representations. The background connections to quantum optics and parameter differentiation are discussed in [9, 7]; the calculations here require neither the uniqueness theorem for canonical commutation relations nor a formal use of BCH outside its domain.

11.1 Position, momentum, and an exact symmetric factorization

On $L^2(\mathbb{R})$, let $\hbar > 0$ and consider

$$(Q\psi)(x) = x\psi(x), \quad (P\psi)(x) = -i\hbar\psi'(x). \quad (11.1)$$

On the common invariant domain $\mathcal{S}(\mathbb{R})$ of Schwartz functions, direct differentiation gives

$$[Q, P]\psi = i\hbar\psi. \quad (11.2)$$

The self-adjoint multiplication and differentiation operators define, for real a, b , unitary operators

$$(U(a)\psi)(x) = e^{iax}\psi(x), \quad (V(b)\psi)(x) = \psi(x + b\hbar). \quad (11.3)$$

They are e^{iaQ} and e^{ibP} , respectively; the second assertion follows by diagonalizing P with the Fourier transform. Comparing the two actions immediately proves the Weyl relation

$$U(a)V(b) = e^{-iab\hbar}V(b)U(a). \quad (11.4)$$

Define

$$(W(a, b)\psi)(x) = e^{ia(x+b\hbar/2)}\psi(x + b\hbar). \quad (11.5)$$

For fixed a, b , the operators $W(ta, tb)$ form a strongly continuous one-parameter unitary group: substitution in [equation \(11.5\)](#) verifies the group law, and continuity follows first on smooth compactly supported functions and then by density. Its derivative at zero on $\mathcal{S}(\mathbb{R})$ is $i(aQ + bP)$. This determines the exponential of the self-adjoint closure of $aQ + bP$. One can also verify essential self-adjointness directly: for $b \neq 0$, multiplication by $\exp(-iax^2/(2b\hbar))$ conjugates bP to $aQ + bP$ and preserves Schwartz space. The case $b = 0$ is multiplication. Accordingly,

$$W(a, b) = e^{\overline{iaQ + bP}}. \quad (11.6)$$

Here and below an exponential of a real linear combination denotes this self-adjoint closure.

Theorem 11.1 (Canonical BCH identities). *For all real a, b ,*

$$e^{iaQ}e^{ibP} = e^{-iab\hbar/2}e^{i(aQ+bP)} = \exp\left(i\left(aQ + bP - \frac{1}{2}ab\hbar I\right)\right), \quad (11.7)$$

$$e^{i(aQ+bP)} = e^{iaQ/2}e^{ibP}e^{iaQ/2}. \quad (11.8)$$

Proof. The product in the first line acts as $e^{iax}\psi(x+b\hbar)$; comparison with [equation \(11.5\)](#) gives the scalar phase. The symmetric product acts as $e^{iax/2}e^{ia(x+b\hbar)/2}\psi(x+b\hbar)$, which is exactly [equation \(11.5\)](#). Both proofs hold on all of L^2 , since only bounded unitary operators occur in the final identities. $\square$

Thus the second-order symmetric splitting becomes exact for this central commutator. More generally, for any finite sequence of real pairs,

$$\prod_{j=1}^m W(a_j, b_j) = \exp\left(-\frac{i\hbar}{2} \sum_{j < k} (a_j b_k - b_j a_k)\right) W\left(\sum_j a_j, \sum_j b_j\right). \quad (11.9)$$

For two pairs this follows by composing [equation \(11.5\)](#); induction gives the displayed sum. It is the full central-commutator expansion: all commutators of length three and above vanish.

11.2 Why a commutator on a dense domain is not enough

The preceding explicit proof is essential to the scope of the result. On $L^2(0, 1)$ let Q again be multiplication by x , and let $P = -i\hbar d/dx$ have periodic self-adjoint boundary conditions. On the dense common invariant domain $C_c^\infty(0, 1)$ one has $[Q, P] = i\hbar I$. Nevertheless $e^{iP} = I$, since the eigenvalues of P are $2\pi n$. A Weyl relation inferred solely from this commutator would give

$$e^{iaQ} = e^{-ia} e^{iaQ}, \quad (11.10)$$

which is false for general a . The failure is a domain/integrability issue, not an error in the formal central-commutator calculation.

There cannot instead be a pair of bounded operators in a unital Banach algebra satisfying $[A, B] = I$. Indeed, the derivation rule gives $[A, B^n] = nB^{n-1}$. No power of B can vanish, since its least vanishing power would contradict that identity. Therefore

$$n \|B^{n-1}\| \leq 2 \|A\| \|B^n\| \leq 2 \|A\| \|B\| \|B^{n-1}\| \quad (11.11)$$

implies $n \leq 2 \|A\| \|B\|$ for every positive integer n , an impossibility. Finite-dimensional matrices are ruled out even more quickly by taking traces. These observations explain why the bounded-operator BCH theorem cannot be applied directly to Q, P .

11.3 Bargmann–Fock realization and normal ordering

Consider the Hilbert space of entire functions with

$$\|f\|^2 = \frac{1}{\pi} \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} d^2z < \infty. \quad (11.12)$$

The functions $e_n(z) = z^n / \sqrt{n!}$ form an orthonormal basis. Orthogonality and norms follow by polar integration; expanding an entire function in its Taylor series and integrating first on circles gives $\|\sum c_n z^n\|^2 = \sum n! |c_n|^2$, proving completeness as well. On the dense polynomial domain define

$$af = f', \quad a^\dagger f = zf. \quad (11.13)$$

Then $ae_n = \sqrt{n} e_{n-1}$, $a^\dagger e_n = \sqrt{n+1} e_{n+1}$, and

$$[a, a^\dagger] = I, \quad [a^\dagger, a] = -I. \quad (11.14)$$

The notation $\dagger$ is justified by these matrix elements and the usual closed adjoint realizations.

For $v \in \mathbb{C}$ define a bounded operator by its action

$$(D(v)f)(z) = e^{vz - |v|^2/2} f(z - \bar{v}). \quad (11.15)$$

A change of variables $z = w + \bar{v}$ gives the exact weight identity

$$\left| e^{vz - |v|^2/2} \right|^2 e^{-|z|^2} = e^{-|w|^2}. \quad (11.16)$$

Thus $D(v)$ is an isometry, with inverse $D(-v)$, hence unitary. Strong continuity follows on the dense set of polynomials from the formula and a Gaussian integrable majorant locally in v , and then on the whole space by unitarity. The family $D(tv)$, $t \in \mathbb{R}$, has derivative $va^\dagger - \bar{v}a$ on polynomials and obeys its one-parameter group law. Consequently

$$D(v) = \exp(\overline{va^\dagger - \bar{v}a}), \quad (11.17)$$

where the outer bar denotes operator closure, not complex conjugation. To justify that it is the closure, rather than an unspecified extension, note that every polynomial is an analytic vector for this skew-symmetric operator. For a polynomial of degree at most m , its k th power has norm at most $C(2|v|)^k \sqrt{(m+k)!}$, by the raising/lowering rules. Division by $k!$ makes the exponential series convergent for every fixed parameter. Here is a direct justification of the core assertion. Write $T = va^\dagger - \bar{v}a$ on polynomials and let G be the skew-adjoint generator of $D(tv)$, so $T \subset G$. The same bounds for T^{k+1} allow termwise differentiation of $\sum t^k T^k p / k!$ in the graph norm of G ; uniqueness of the group differential equation identifies this series with $D(tv)p$. If a vector f is orthogonal to $(\lambda I - T)$ applied to all polynomials, for $\lambda = 1$ or -1 , then $\langle f, T^k p \rangle = \lambda^k \langle f, p \rangle$ and hence $\langle f, D(tv)p \rangle = e^{\lambda t} \langle f, p \rangle$. The left side is bounded for all real t , so $\langle f, p \rangle = 0$ and $f = 0$. Thus both $(I - T)$ and $(I + T)$ have dense range. For the closure $\bar{T}$, skew symmetry gives $\|(I \pm \bar{T})x\|^2 = \|x\|^2 + \|\bar{T}x\|^2$, so these ranges are also closed and therefore surjective. Since $\bar{T} \subset G$ and $I - \bar{T}$ is surjective, for any $y \in D(G)$ choose $x \in D(\bar{T})$ with $(I - \bar{T})x = (I - G)y$. Then $(I - G)(y - x) = 0$; skew adjointness of G implies that $I - G$ is injective, so $y = x$. Thus $\bar{T} = G$, establishing the stated closure.

Theorem 11.2 (Normal ordering and composition). *On polynomials, with bounded extension understood for the whole product,*

$$D(v) = e^{va^\dagger} e^{-\bar{v}a} e^{-|v|^2/2}. \quad (11.18)$$

As an identity of bounded operators on the entire Fock space,

$$D(v)D(u) = D(v+u) \exp\left(\frac{v\bar{u} - u\bar{v}}{2}\right). \quad (11.19)$$

Proof. On a polynomial, the Taylor sum for $e^{-\bar{v}a}$ is finite and gives translation by $-\bar{v}$. The exponential of $va^\dagger$ gives multiplication by e^{vz} ; its series converges in Fock norm on any polynomial. Thus the right side of [equation \(11.18\)](#) is exactly [equation \(11.15\)](#) on a dense domain, and has the stated unitary extension. This assertion does not declare its two individual unbounded factors to be everywhere defined.

To prove composition, apply the action twice. The multiplier is $\exp(vz - |v|^2/2 + u(z - \bar{v}) - |u|^2/2)$, and the argument is $z - \bar{v} - \bar{u}$. Dividing by the multiplier for $D(v+u)$ leaves exactly the scalar in [equation \(11.19\)](#). Since its exponent is purely imaginary, it is a phase. This also proves the one-parameter group law used above by taking u, v to be real multiples of the same complex number. $\square$

Equivalently, with $A(v) = va^\dagger - \bar{v}a$, $[A(v), A(u)] = (v\bar{u} - u\bar{v})I$ on the polynomial domain. This explains the central BCH phase but is not needed to justify the bounded identity. Iterating it gives the complete finite-product formula

$$\prod_{j=1}^m D(v_j) = \exp\left(\frac{1}{2} \sum_{j < k} (v_j \bar{v}_k - v_k \bar{v}_j)\right) D\left(\sum_j v_j\right). \quad (11.20)$$

11.4 Full state and matrix-element expansions

Writing $|n\rangle = e_n$, the vacuum formula is the convergent all-terms expansion

$$|v\rangle := D(v)|0\rangle = e^{-|v|^2/2} \sum_{n=0}^{\infty} \frac{v^n}{\sqrt{n!}} |n\rangle, \quad \| |v\rangle \| = 1. \quad (11.21)$$

The norm follows by summing the scalar exponential, and taking the inner product gives

$$\langle u | v \rangle = \exp\left(-\frac{|u|^2 + |v|^2}{2} + \bar{u}v\right). \quad (11.22)$$

There is also an exact finite coefficient formula for every matrix element:

$$\langle n | D(v) | m \rangle = e^{-|v|^2/2} \sqrt{n!m!} \sum_{k=0}^{\min(n,m)} \frac{v^{n-k} (-\bar{v})^{m-k}}{(n-k)!(m-k)!k!}. \quad (11.23)$$

Indeed, in the normal-ordered expression first lower m to k , producing $(-\bar{v})^{m-k} \sqrt{m!/k!}/(m-k)!$, and then raise k to n , producing $v^{n-k} \sqrt{n!/k!}/(n-k)!$. Summing over the possible intermediate levels proves the formula. For $n \geq m$ it can equivalently be written

$$\langle n | D(v) | m \rangle = e^{-|v|^2/2} \sqrt{\frac{m!}{n!}} v^{n-m} L_m^{(n-m)}(|v|^2), \quad (11.24)$$

where the polynomial is defined, with no additional identity assumed, by $L_m^{(\alpha)}(x) = \sum_{j=0}^m \binom{m+\alpha}{m-j} (-x)^j / j!$. Substituting $j = m - k$ in [equation \(11.23\)](#) proves the equivalence. The case $m \geq n$ follows in the same way or by adjunction with $D(v)^* = D(-v)$.

Finally, finitely many independent modes are handled by tensoring these representations. All cross-mode commutators vanish; consequently every scalar product $v\bar{u}$ in the phase formulas is replaced by $\sum_{\ell} v_{\ell} \bar{u}_{\ell}$, and the vacuum and matrix-element formulas factor by mode. No further BCH terms appear. The resulting central-extension multiplication is the Heisenberg group law, and its commutator subgroup is central, explicitly exhibiting its two-step nilpotence.

12 Exact computation, additional terms, and reproducibility

The all-order assertions in this paper are proved algebraically or analytically; a finite computation is not their substitute. Computation is useful for certifying transcription, signs, bracket conventions, and the additional explicit terms. The accompanying implementation uses only Python's standard library and exact rational arithmetic.

12.1 Representation and independently organized calculations

An associative polynomial is stored as a dictionary from words in $\{X, Y\}$ to rational coefficients. The empty word represents 1. Multiplication concatenates words and discards degrees beyond a requested bound. Exponential and logarithm are then evaluated by their defining series, which become finite under this truncation. Because no relation is imposed on the two generators, equality of the resulting dictionaries is equality in the truncated free associative algebra, not a test on selected numerical matrices.

The implementation code `verify_bch.py` performs the following distinct constructions. The reference BCH expansion is the ordinary logarithm of the product of the two exponential series. The word-cut formula is evaluated by a dynamic program over all contiguous cuts of each word, including words with zero final coefficient. Independently, the Poincaré expansion is assembled by repeated application of $e^{\text{ad}_X} e^{t \text{ad}_Y} - I$ and exact integration of powers of t . The Dynkin projection, the Bernoulli linear part, and the quadratic kernel are expanded separately and compared with the same associative coefficients. Finally the primitive-element condition is checked by explicitly computing the reduced unshuffle coproduct.

Zassenhaus exponents are calculated twice: by residual-product extraction [equation \(8.3\)](#) and by the finite Lie recurrence [equation \(8.9\)](#). The ordered product is then reconstructed. The source article’s displayed degree-one through degree-six BCH expressions and degree-two through degree-four Zassenhaus expressions are entered independently as nested brackets and compared coefficient by coefficient. The symmetric splitting is checked for oddness and for its cubic coefficient.

For the supplied run, all exported expansions are through total degree 12. The independent word-cut, Poincaré, coproduct, quadratic-kernel, Dynkin-projection, and dual Zassenhaus-construction comparisons run through degree 10. Exponential/logarithm reconstruction, all exported commutator-table reconstructions, linear-in- Y coefficients, the interchange parity, the ordered Zassenhaus product, and the symmetric-splitting parity are checked through degree 12. The machine-readable report records **30 passed checks**, including grouped checks over whole degree ranges. These are exact finite algebraic verifications, not floating-point tolerance tests and not a claim of formal proof-assistant certification.

12.2 Two different commutator encodings

The right-nested data use precisely $R(w)$ from [equation \(1.6\)](#):

$$Z_n = \sum_{|w|=n} \frac{c(w)}{n} R(w). \quad (12.1)$$

This encoding is redundant. In particular, every word ending in two equal letters represents zero, and such rows are omitted. Remaining rows are not claimed to be linearly independent.

For compact additional tables, we use the conventional standard Lyndon bracketing. Order the alphabet by $X < Y$ and words lexicographically, with a proper prefix preceding its extension. A nonempty word is Lyndon if it precedes every nonempty proper suffix. Define $L(X) = X$, $L(Y) = Y$; if w has length greater than one, write $w = uv$ with v its longest proper Lyndon suffix and set

$$L(w) = [L(u), L(v)]. \quad (12.2)$$

In the table-generation algorithm this recursion is applied to the resulting shorter words, so every entry is an explicitly specified bracket polynomial. For example,

$$L(XXY) = [X, [X, Y]], \quad L(XYY) = [[X, Y], Y]. \quad (12.3)$$

Thus $L(XYY)$ is *not* $R(XYY)$, which is zero. Confusing the two conventions would corrupt many coefficients.

No theorem about a Lyndon basis is needed to verify these tables. The program selects coefficients by triangular elimination of expanded bracket polynomials and requires the entire residual associative polynomial to be zero. It then reconstructs each exported table once more and compares every coefficient with the original polynomial. A table is therefore a certificate for an explicit finite commutator identity whether or not one invokes the general basis theorem. The counts in [table 5](#) are numbers of nonzero entries in these encodings, not dimensions of homogeneous free Lie spaces.

12.3 Included files and regeneration

The principal data files are summarized in [table 1](#). In an associative file a row means the coefficient of the literal word. In a Lyndon file it means the coefficient multiplying $L(w)$; in the right-commutator file it multiplies $R(w)$. Numerator and denominator are separate integers, and missing rows have coefficient zero. From the archive’s root directory, the supplied data and table appendix can be regenerated with

File in data/	Meaning
bch_associative.csv	Every nonzero BCH word coefficient through degree 12.
bch_right_commutators.csv	Dynkin-projection coefficients, with identically zero right brackets omitted.
bch_lyndon.csv	Compact BCH commutator expansion through degree 12.
zassenhaus_associative.csv	Associative coefficients of each C_n , $2 \leq n \leq 12$.
zassenhaus_lyndon.csv	Compact commutator expansions of the same C_n .
strang_lyndon.csv	Logarithm of $e^{X/2}e^Ye^{X/2}$ through degree 12.
bernoulli_plus.csv	The coefficients $b_n^+ = B_n^+/n!$ through degree 12.
verification_report.json	Parameters, exact check outcomes, counts, and elapsed time.

Table 1: The machine-readable companion expansions.

```
python code/verify_bch.py --degree 12 --check-degree 10
python code/make_tables.py
```

The first command requires Python 3.9 or later and no third-party package. Larger degrees are supported but have rapidly increasing cost because the number of words is exponential. The finite coefficient formulas and recurrences in the paper, rather than the chosen degree limit, specify all orders.

The typeset paper can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error paper.tex
```

or by running `pdflatex paper.tex` repeatedly until references and the table of contents stabilize. A standard TeX installation with the packages listed in the preamble is sufficient. The archive includes the main source and every input file; no external bibliography database or downloaded mathematical source is needed for compilation.

A Free Lie algebras and universal enveloping Hopf algebras

This appendix proves the universal-enveloping statement used by the source article and explains why a calculation in free associative words is a universal Lie identity.

A.1 The universal algebra and its ordered normal forms

For a Lie algebra $\mathfrak{g}$ over a field of characteristic zero, define

$$U(\mathfrak{g}) = T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] : x, y \in \mathfrak{g} \rangle. \quad (\text{A.1})$$

A linear map $\mathfrak{g} \rightarrow A$ to an associative algebra extends uniquely to $T(\mathfrak{g})$ by multiplying images of tensors. It factors through [equation \(A.1\)](#) exactly when it preserves the Lie bracket, with A given its commutator bracket. This proves the universal property directly. It remains important that the map $\mathfrak{g} \rightarrow U(\mathfrak{g})$ be injective.

Theorem A.1 (Ordered-monomial theorem). *Choose a totally ordered basis (e_i) of $\mathfrak{g}$. The monomials $e_{i_1} \cdots e_{i_n}$ with $i_1 \leq \cdots \leq i_n$, together with 1, form a vector-space basis of $U(\mathfrak{g})$. In particular $\mathfrak{g} \rightarrow U(\mathfrak{g})$ is injective.*

Proof. Use the linear rewriting rule

$$e_j e_i \mapsto e_i e_j + [e_j, e_i] \quad (j > i), \quad (\text{A.2})$$

expanding the bracket as a finite linear combination of basis elements. Rewriting terminates: a swapped adjacent inverted pair reduces the inversion number while preserving word length, and a bracket term reduces length. Order these two nonnegative integers lexicographically, with length first; after every replacement each resulting monomial has smaller measure. The multiset extension of this order gives termination for finite linear combinations.

Disjoint replacements commute. The only overlapping ambiguity comes from a word abc with basis indices satisfying $a > b > c$. Reducing the first inverted pair first gives, before normalizing the shorter terms,

$$cba + [b, c]a + b[a, c] + [a, b]c.$$

Reducing the second pair first gives

$$cba + c[a, b] + [a, c]b + a[b, c].$$

Their difference reduces, using rules on shorter products, to

$$[[b, c], a] + [b, [a, c]] + [[a, b], c] = [[b, c], a] + [[c, a], b] + [[a, b], c] = 0$$

by Jacobi. Thus every possible first conflict can be resolved. Induction on the terminating measure now proves uniqueness of the reduced form: compare the first two reductions, resolve a disjoint or overlapping conflict as above, and use the induction hypothesis on all terms below the initial measure. This is the usual confluence argument written here for this specific system.

Hence every tensor has a unique normal form in the span of ordered words. Each relation in [equation \(A.1\)](#), also when multiplied on either side by words, has zero normal form, because its two sides are a permitted local rewrite. Therefore the whole defining ideal has zero normal form. Conversely, reduction changes a tensor only by elements of that ideal. The quotient is consequently identified with the vector space of ordered normal forms, proving both spanning and linear independence. In degree one the basis elements remain distinct, proving injectivity. $\square$

The proof uses the basic bracket axioms, not a prior matrix representation of $\mathfrak{g}$.

A.2 The free Lie algebra and the free associative algebra

Let $F(X, Y)$ be the abstract free Lie algebra, constructed from bracketed words by imposing bilinearity, skew symmetry, and Jacobi. Giving a Lie homomorphism from $F(X, Y)$ is exactly the same as choosing two elements in the target Lie algebra: substitution defines the map, and the imposed relations are precisely what makes it well-defined.

The universal property of $U(F(X, Y))$ therefore says that an associative homomorphism from it to an arbitrary associative algebra A is exactly a choice of two elements of A . The free associative algebra $\mathbb{K}\langle X, Y \rangle$ has the same property. More explicitly, sending its two generators to the corresponding elements of $U(F(X, Y))$ defines one map; the free Lie map $F(X, Y) \rightarrow \mathbb{K}\langle X, Y \rangle$ followed by the universal property of U defines the other. Both compositions fix the generators, and hence are identities. Thus

$$U(F(X, Y)) \cong \mathbb{K}\langle X, Y \rangle. \quad (\text{A.3})$$

By [theorem A.1](#), $F(X, Y)$ injects into this algebra, and its image is exactly the Lie subalgebra generated by X, Y . This identifies the concrete $\mathcal{L}(X, Y)$ used throughout the paper with the abstract free Lie algebra. In particular, an equality of its bracket polynomials verified as an associative-word equality is an identity valid under substitution in *every* Lie algebra.

A.3 The enveloping Hopf structure

Define on $\mathfrak{g}$ the maps

$$\Delta x = x \otimes 1 + 1 \otimes x, \quad \varepsilon(x) = 0, \quad S(x) = -x. \quad (\text{A.4})$$

Extend Δ, ε as algebra homomorphisms and S as an antihomomorphism. These definitions respect the relations in [equation \(A.1\)](#): for Δ , the mixed tensor-factor terms in $[\Delta x, \Delta y]$ cancel, leaving $\Delta[x, y]$; for S , applying it to $xy - yx - [x, y]$ gives the negative of the same relation. Coassociativity and the counit identities hold on generators and hence on all of $U(\mathfrak{g})$. The antipode identities follow from them on generators and induction on products. Thus $U(\mathfrak{g})$ is a Hopf algebra.

Under [equation \(A.3\)](#), this is exactly the unshuffle Hopf algebra of [section 2](#); degree completion gives its formal noncommutative power-series version. For a general $\mathfrak{g}$, ordinary tensor length need not be a grading of $U(\mathfrak{g})$, since a bracket relation changes length. Formal exponentials in that setting can instead be taken in $U(\mathfrak{g})[[t]]$ with tX, tY as arguments. This avoids assuming an inappropriate grading while preserving every coefficient identity.

B Coefficient certificates and additional complete degrees

B.1 All nonzero associative coefficients in degrees five and six

[Table 2](#) is a finite certificate for [equations \(3.15\) and \(3.16\)](#). Its two coefficient columns on each side come from the cut formula [equation \(3.4\)](#) and expansion of the displayed right-nested brackets, respectively. Both calculations give zero on every word absent from the table. For degree five these are only $XXXXX$ and $YYYYY$; for degree six all 64 possible words, including the 36 zero coefficients, are compared by the accompanying script. Thus equality is checked in the free associative algebra, not in a matrix quotient that might hide a Lie identity.

Table 2: Complete nonzero word-coefficient certificates for the displayed fifth and sixth degrees.

Z_5			Z_6		
w	c_{cut}	c_{bracket}	w	c_{cut}	c_{bracket}
XXXXY	$-\frac{1}{720}$	$-\frac{1}{720}$	XXXXYY	$-\frac{1}{1440}$	$-\frac{1}{1440}$
XXXYY	$\frac{1}{180}$	$\frac{1}{180}$	XXXYYX	$\frac{1}{360}$	$\frac{1}{360}$
XXXYY	$\frac{1}{180}$	$\frac{1}{180}$	XXXYYY	$\frac{1}{360}$	$\frac{1}{360}$
XXYXX	$-\frac{1}{120}$	$-\frac{1}{120}$	XXYXXY	$-\frac{1}{240}$	$-\frac{1}{240}$
XXYXY	$-\frac{1}{120}$	$-\frac{1}{120}$	XXYXYX	$-\frac{1}{240}$	$-\frac{1}{240}$
XXYYY	$-\frac{1}{120}$	$-\frac{1}{120}$	XXYYYX	$-\frac{1}{240}$	$-\frac{1}{240}$
XXYYY	$\frac{1}{180}$	$\frac{1}{180}$	XXYYYY	$-\frac{1}{1440}$	$-\frac{1}{1440}$
XYXXX	$\frac{1}{180}$	$\frac{1}{180}$	XYXXX	$\frac{1}{360}$	$\frac{1}{360}$
XYXXX	$-\frac{1}{120}$	$-\frac{1}{120}$	XYXXX	$-\frac{1}{240}$	$-\frac{1}{240}$
XYXXY	$\frac{1}{30}$	$\frac{1}{30}$	XYXXY	$\frac{1}{60}$	$\frac{1}{60}$
XYXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	XYXXYY	$\frac{1}{360}$	$\frac{1}{360}$
XYYYX	$-\frac{1}{120}$	$-\frac{1}{120}$	XYYYXX	$-\frac{1}{240}$	$-\frac{1}{240}$
XYYYX	$-\frac{1}{120}$	$-\frac{1}{120}$	XYYYXY	$-\frac{1}{240}$	$-\frac{1}{240}$
XYYYX	$\frac{1}{180}$	$\frac{1}{180}$	XYYYXY	$\frac{1}{360}$	$\frac{1}{360}$
XYYYY	$-\frac{1}{720}$	$-\frac{1}{720}$	YXXXXX	$-\frac{1}{360}$	$-\frac{1}{360}$
YXXXX	$-\frac{1}{720}$	$-\frac{1}{720}$	YXXXXX	$\frac{1}{240}$	$\frac{1}{240}$
YXXXX	$\frac{1}{180}$	$\frac{1}{180}$	YXXXXY	$\frac{1}{240}$	$\frac{1}{240}$
YXXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	YXXXYX	$-\frac{1}{360}$	$-\frac{1}{360}$
YXXYY	$-\frac{1}{120}$	$-\frac{1}{120}$	YXXYYX	$-\frac{1}{60}$	$-\frac{1}{60}$
YXXYY	$-\frac{1}{120}$	$-\frac{1}{120}$	YXXYYX	$\frac{1}{240}$	$\frac{1}{240}$
YXXYY	$\frac{1}{30}$	$\frac{1}{30}$	YXXYYY	$-\frac{1}{360}$	$-\frac{1}{360}$
YXXYY	$-\frac{1}{120}$	$-\frac{1}{120}$	YYXXXX	$\frac{1}{1440}$	$\frac{1}{1440}$
YXXYY	$\frac{1}{180}$	$\frac{1}{180}$	YYXXXY	$\frac{1}{240}$	$\frac{1}{240}$
YYXXX	$\frac{1}{180}$	$\frac{1}{180}$	YYXXYY	$\frac{1}{240}$	$\frac{1}{240}$
YYXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	YYXXYY	$\frac{1}{240}$	$\frac{1}{240}$
YYXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	YYXXYY	$-\frac{1}{360}$	$-\frac{1}{360}$
YYXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	YYXXYY	$-\frac{1}{360}$	$-\frac{1}{360}$
YYXXY	$-\frac{1}{120}$	$-\frac{1}{120}$	YYXXYY	$-\frac{1}{360}$	$-\frac{1}{360}$
YYXXY	$\frac{1}{180}$	$\frac{1}{180}$	YYXXYY	$\frac{1}{1440}$	$\frac{1}{1440}$
YYXXY	$\frac{1}{180}$	$\frac{1}{180}$	—	—	—
YYXXY	$-\frac{1}{720}$	$-\frac{1}{720}$	—	—	—

B.2 Complete seventh and eighth BCH degrees

In the following table the entries specify exactly $Z_n = \sum_w c_w L(w)$ for $n = 7, 8$, using [equation \(12.2\)](#); there are no omitted nonzero terms in this convention. Expanding the brackets and applying [equation \(3.4\)](#) proves each identity by the same finite procedure as in the preceding table. The CSV file continues these full expansions through degree twelve.

Table 3: Additional complete BCH homogeneous polynomials in standard Lyndon bracketing.

Z_7		Z_8	
w	coefficient of $L(w)$	w	coefficient of $L(w)$
XXXXXXY	$\frac{1}{30240}$	XXXXXXYY	$\frac{1}{60480}$
XXXXXY	$-\frac{1}{5040}$	XXXXXYXY	$-\frac{1}{15120}$
XXXXYXY	$\frac{1}{10080}$	XXXXXYYY	$-\frac{1}{10080}$
XXXXYYY	$\frac{1}{3780}$	XXXXYXXY	$\frac{1}{20160}$
XXXYXXY	$\frac{1}{10080}$	XXXYXYXY	$-\frac{1}{20160}$
XXXYXY	$\frac{1}{1680}$	XXXYYYXY	$\frac{1}{2520}$
XXXYXXY	$\frac{1}{1260}$	XXXYYYYY	$\frac{23}{120960}$
XXXYYYY	$\frac{1}{3780}$	XXXYXXYY	$\frac{1}{4032}$
XXYXXYY	$\frac{1}{2016}$	XXYXYXY	$-\frac{1}{10080}$
XXYXYXY	$-\frac{1}{5040}$	XXYXYYY	$\frac{13}{30240}$
XXYXYYY	$\frac{13}{15120}$	XXYXXYY	$\frac{1}{20160}$
XXYXXYY	$\frac{1}{10080}$	XXYYYXY	$-\frac{1}{3024}$
XXYYYXY	$-\frac{1}{1512}$	XXYYYYY	$-\frac{1}{10080}$
XXYYYYY	$-\frac{1}{5040}$	XXYXXYY	$\frac{1}{2520}$
XYXXXY	$\frac{1}{1260}$	XXYXYYY	$-\frac{1}{4032}$
XYXYYY	$-\frac{1}{2016}$	XXYXXYY	$-\frac{1}{10080}$
XYXXYYY	$-\frac{1}{5040}$	XXYYYYY	$\frac{1}{60480}$
XYYYYY	$\frac{1}{30240}$	—	—

B.3 Complete fifth and sixth Zassenhaus exponents

These are the next two full exponents after the terms displayed on the source page. A coefficient multiplies $L(w)$, not $R(w)$. Every entry follows either from [equation \(8.3\)](#) or from [equation \(8.9\)](#); the two constructions agree exactly. Higher exponents through degree twelve are supplied as CSV data.

Table 4: Complete fifth and sixth Zassenhaus exponents.

C_5		C_6	
w	coefficient of $L(w)$	w	coefficient of $L(w)$
XXXXY	$\frac{1}{120}$	XXXXXY	$-\frac{1}{720}$
XXXYY	$-\frac{1}{30}$	XXXXY	$\frac{1}{144}$
XXYXY	$-\frac{1}{60}$	XXXYYY	$-\frac{1}{72}$
XXYYY	$\frac{1}{20}$	XXYYYY	$\frac{1}{72}$
XYXY	$-\frac{1}{20}$	XYXYYY	$-\frac{1}{72}$
XYYYY	$-\frac{1}{30}$	XYYYYY	$-\frac{1}{144}$

B.4 Sizes of the exported expansions

Degree	BCH words	BCH Lyndon entries	Zassenhaus Lyndon entries
1	2	2	0
2	2	1	1
3	6	2	2
4	4	1	3
5	30	6	6
6	28	5	6
7	126	18	18
8	124	17	29
9	390	55	56
10	388	55	95
11	2046	186	186
12	2044	185	326

Table 5: Numbers of nonzero coefficients in the supplied encodings. The degree-one Zassenhaus factor e^Y is separated from the exponents C_n , $n \geq 2$.

C Concordance with the source article

This concordance refers to the fixed revision cited in [1]. It tracks mathematical content rather than historical prose. Repeated formulas on the page are grouped, but different hypotheses and convergence interpretations are distinguished. An entry marked “corrected” gives the valid statement and explains why the unqualified original cannot be asserted. Theorems appearing only as names in links are not statements made by that page; the Trotter and higher-order splitting results are nevertheless included as useful additional consequences.

Table 6: Claim-by-claim mathematical coverage.

Source location	Formula or assertion	Proof or qualification here
Introduction	Exponential and logarithm series; $e^X e^Y = e^Z$ as a universal Lie series.	Theorems 1.1, 2.4 and 6.1.
Introduction	Commutator interpretation and Jacobi identity.	Equations (1.3) and (1.4); associative expansion.
Introduction; homomorphism discussion	Local exponential chart; local multiplication determined by the Lie bracket; integration of homomorphisms.	Section 6.4 and theorem 6.4; global hypotheses stated.
Dynkin formula	Complete block sum, factorial denominators, total-degree denominator, right nesting.	Theorem 3.2 and equation (3.6).
Dynkin formula	Last-block vanishing conditions.	Proof following equation (3.7).
Dynkin formula	Every displayed BCH term through degree six.	Equations (3.11) to (3.16); table 2.
Dynkin formula	All nonlinear terms belong to the ideal generated by $[X, Y]$.	Theorem 3.3.
Dynkin formula	$Z(Y, X) = -Z(-X, -Y)$ and homogeneous interchange signs.	Theorem 3.4.
Integral form	Poincaré integral and the function ψ .	Theorem 4.3 and equation (4.14).
Matrix form	Full associative block expansion and terms through degree four.	Equations (3.2) and (3.17) to (3.19).
Matrix form; note 2	Conditions involving $\log 2$, including a separately given Z .	Theorem 6.1 and the discussion following its proof.
Matrix form	Trace additivity.	Theorem 6.2; arbitrary logarithm branches treated separately.
Convergence issues	Traceless 2×2 example and failure of convergence.	Theorem 6.3; all logarithms classified, including conditional-series obstruction.
Convergence issues	Nonisomorphic global groups may have the same Lie algebra.	The real line and circle example in section 6.4.
Convergence issues	Submultiplicative-norm bound $(\log 2)/2$.	Equation (6.4); distinguished from grouped BCH convergence.
Special cases	Commuting variables.	Equation (7.1).
Special cases; existence proof	Central-commutator truncation and the disentangling identity without smallness.	Theorem 7.1, including its differential-equation proof.
Special cases	Linear-in- Y formula, hyperbolic cotangent, Bernoulli coefficients.	Equation (5.4) and theorem 4.2; all higher Y orders in equations (5.2) and (5.3).
Special cases	$[X, Y] = sY$: exact summation, excluded poles, removable zero.	Theorem 7.2; Taylor convergence versus global resummation explained.
Special cases	Eigenvector braiding and adjoint dilation.	Equation (7.9) and theorem 4.1.
Existence results	Formal Hopf algebra and primitive/group-like equivalence.	Theorem 2.1; the tensor product is completed.
Existence results	Group-like elements form a group; primitive elements are precisely Lie series.	Theorems 2.1 and 2.3.

Table 6 (continued)

Source location	Formula or assertion	Proof or qualification here
Existence results	The free associative algebra is the universal enveloping algebra of the free Lie algebra.	Equation (A.3) and theorem A.1; compatible Hopf structure in equation (A.4).
Existence results	Lie-series existence independently of knowing all coefficients.	Theorem 2.4; independent differential recurrence equation (4.19).
Zassenhaus formula	Ordered product and every displayed C_2, C_3, C_4 term.	Theorem 8.1 and equations (8.10), (8.11) and (8.13).
Zassenhaus formula	Compact product after left multiplication by e^{-tX} .	Equation (8.1) with $C_1 = Y$, as explained in its proof.
Zassenhaus formula	All subsequent exponents and analytic meaning.	Equations (8.3) and (8.9) and theorem 8.2; table 4.
Suzuki–Trotter discussion	Product approximations and higher-order factorizations.	Section 9, including full effective-exponent formulas.
Campbell identity	$\text{Ad}_{e^X} = e^{\text{ad}_X}$; complete nested-commutator expansion and ODE.	Theorem 4.1 and equation (4.4).
Campbell identity	General conjugation/braiding and its central reduction.	Equations (4.2), (4.3) and (7.2).
Infinitesimal version	Full series for $e^{-X}de^X$.	Equations (4.6) and (10.2).
Infinitesimal version	Structure-constant expansion; matrix M and W .	Equations (10.3) and (10.4); all orders contracted explicitly.
Infinitesimal version	Quotient by M ; identification of a vielbein/coframe.	Corrected by equations (10.6) to (10.8): the coframe matrix is W , not M .
Infinitesimal version	Pullback metric and Killing form.	Corrected in equations (10.10) to (10.12); Jacobian and nondegeneracy required.
Quantum mechanics	Canonical commutator and exponentiated formula.	Equation (11.2) and theorem 11.1; concrete operator domains supplied.
Quantum mechanics	Exact symmetric product.	Equation (11.8).
Quantum mechanics	Creation/annihilation commutator; normal ordering.	Equation (11.14) and theorem 11.2.
Quantum mechanics	Displacement composition, phase, Heisenberg nilpotence.	Equations (11.19) and (11.20); section 6.5.
Mathematical note 1	$\psi(e^Y)$ and the $B_1 = +1/2$ convention.	Equations (4.10), (4.11) and (4.15).

Beyond the source’s finite displays. The paper supplies an exact formula for every associative-word coefficient, Dynkin’s complete commutator sum, two total-order constructions, every homogeneous order in the number of Y ’s, an explicit all- X quadratic kernel, every Zassenhaus exponent, full effective exponents for splitting products, all-order coframe and metric expansions, and exact displacement matrix elements. The accompanying tables are finite illustrations of these unlimited formulas, not their claimed endpoint.

References

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